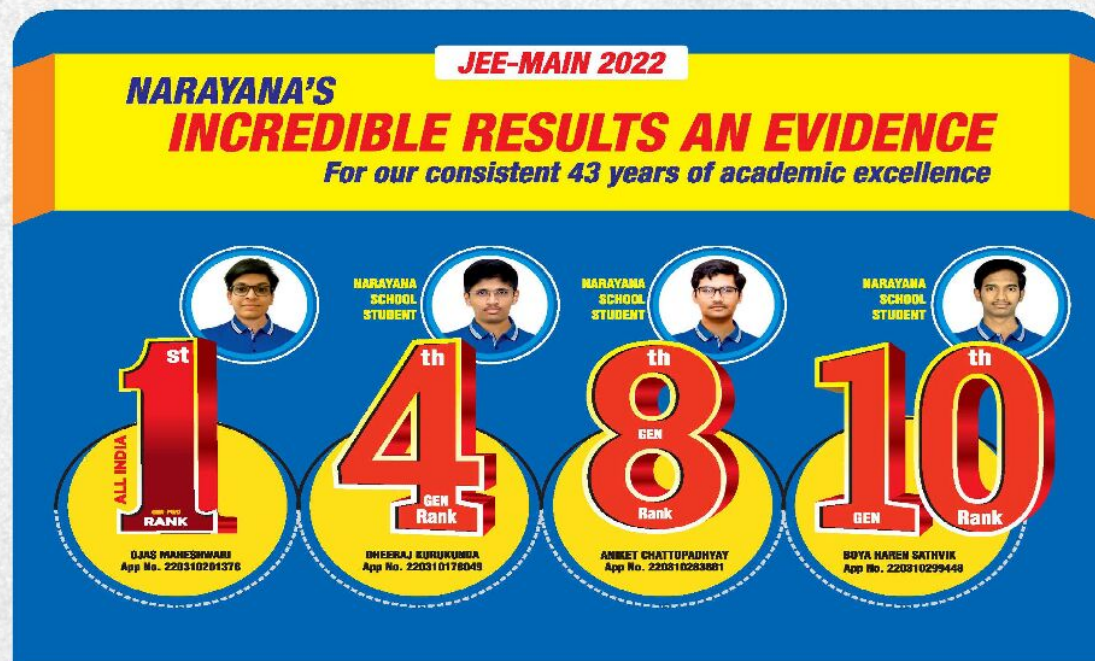




NARAYANA
IIT-JEE ACADEMY - INDIA

NARAYANA GRABS THE LION'S SHARE IN JEE-ADV.2022
EXCELLENCE IS OUR TRADITION 3402 RANKS OVERALL!



MATHS

DIFFERENTIAL CALCULUS
JEE ADVANCED IMPORTANT QUESTIONS

- The value of $\lim_{x \rightarrow 1} \left(\frac{100}{1-x^{100}} - \frac{50}{1-x^{50}} \right)$ is _____
- Let $S_n = 1 + 2 + 3 + \dots + n$ and $P_n = \frac{S_2}{S_2-1} \cdot \frac{S_3}{S_3-1} \cdot \frac{S_4}{S_4-1} \dots \frac{S_n}{S_n-1}$, Where $n \in \mathbb{N}$, ($n \geq 2$)
Then $\lim_{n \rightarrow \infty} P_n =$ _____
- The value of $\lim_{n \rightarrow \infty} \left(\sqrt[3]{(n+1)^2} - \sqrt[3]{(n-1)^2} \right)$ is _____
- The reciprocal of the value of $\lim_{n \rightarrow \infty} \left(1 - \frac{1}{2^2} \right) \left(1 - \frac{1}{3^2} \right) \left(1 - \frac{1}{4^2} \right) \dots \left(1 - \frac{1}{n^2} \right)$ is _____
- The largest value of the non – negative integer a for which $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$ is _____.
- A nonzero polynomial with real coefficients has the property that $f(x) = f'(x) \cdot f''(x)$. If a is the leading coefficient of $f(x)$, then the value of $1/a$ is _____
- The slope of the tangent to the curve $(y - x^5)^2 = x(1 + x^2)^2$ at the point (1, 3) is _____
- Let $f(x) = x + \frac{1}{2x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}}$
Then the value of $f(50) \cdot f'(50)$ is _____
- Let $f(x) = \begin{cases} \frac{x}{2} - 1, & 0 \leq x < 1 \\ \frac{1}{2}, & 1 \leq x \leq 2 \end{cases}$ and $g(x) = (2x+1)(x-k)+3$, $x \geq 0$. Then $g(f(x))$ is continuous at $x = 1$ if $12k$ is equal to _____.
- A differentiable function f is satisfying the relation
 $f(x+y) = f(x) + f(y) + 2xy(x+y) - \frac{1}{3} \forall x, y \in \mathbb{R}$ and
 $\lim_{h \rightarrow 0} \frac{3f(h)-1}{6h} = \frac{2}{3}$. Then the value of $[f(2)]$ is (where $[x]$ represents the greatest integer function) _____.

11. The least integral value of p for which $f''(x)$ is everywhere continuous where

$$f(x) = \begin{cases} x^p \sin\left(\frac{1}{x}\right) + x|x|, & x \neq 0 \\ 0, & x = 0 \end{cases}$$

12. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ and $g: \mathbb{R} \rightarrow \mathbb{R}$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h: \mathbb{R} \rightarrow \mathbb{R}$ by

$$h(x) = \begin{cases} \max\{f(x), g(x)\}, & \text{if } x \leq 0 \\ \min\{f(x), g(x)\}, & \text{if } x > 0 \end{cases}$$

Then, the number of points at which $h(x)$ is not differentiable is

13. Number of points of discontinuity of $f(x) = [x^3 + 3x^2 + 7x + 2]$, where $[.]$ represents the greatest integer function in $[0, 1]$ is _____
14. There is a point (p, q) on the graph of $f(x) = x^2$ and a point (r, s) on the graph of $g(x) = \frac{-8}{x}$, where $p > 0$ and $r > 0$. If the line through (p, q) and (r, s) is also tangent to both the curves at these points, respectively, then the value of $p + r$ is _____
15. A curve is defined parametrically by the equation $x = t^2$ and $y = t^3$. A variable pair of perpendicular lines through the origin O meet the curve at P and Q . If the locus of the point of intersection of the tangents at P and Q is $ay^2 = bx - 1$, then the value of $(a + b)$ is _____.
16. Let $f(x)$ be a non-constant thrice differentiable function defined on $(-\infty, \infty)$ such that $f(x) = f(6 - x)$ and $f'(0) = 0 = f'(2) = f'(5)$. If n is the minimum number of roots of $(f''(x)^2) + f'(x)f'''(x) = 0$ in the interval $[0, 6]$, then the value of n is _____.
17. Let $f(x) = 30 - 2x - x^3$, then the number of positive integral values of x which satisfies $f(f(f(x))) > f(f(-x))$ is _____
18. Consider $f(x) = \frac{1}{1+|x|} + \frac{1}{1+|x-1|}$. Let x_1 and x_2 be points where $f(x)$ attains local minimum and global maximum respectively. If $k = f(x_1) + f(x_2)$, then $6k - 9 =$ _____.
19. A cylindrical container is to be made from certain solid material with the following constraints: It has fixed inner volume of $V \text{ mm}^3$, has a 2 mm thick solid wall and is open at the top. The bottom of the container is solid circular disc of thickness 2 mm and is of radius

equal to the outer radius of the container.

If the volume of the material used to make the container is minimum when the inner radius of the container is 10 mm, then the value of $\frac{V}{250\pi}$ is _____

20. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = |x| + |x^2 - 1|$. The total number of points at which f attains either a local maximum or a local minimum is _____.
21. Let $f(x)$ be a polynomial of degree 5 such that $f(|x|) = 0$ has 8 real distinct roots. Then number of real roots of $f(x) = 0$ is _____
22. The number of integral values of x for which the function $\sqrt{\sin x + \cos x} + \sqrt{7x - x^2 - 6}$ is defined is _____.

23. Match the following lists:

	Function		Type of function
a.	$f(x) = \{(\operatorname{sgn} x)^{\operatorname{sgn} x}\}^n; x \neq 0, n$ is an odd integer	p.	Odd function
b.	$f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$	q.	Even function
c.	$f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ 1, & \text{if } x \text{ is irrational} \end{cases}$	r.	Neither odd nor even function
d.	$f(x) = \max\{\tan x, \cot x\}$	s.	Periodic

24. Consider $\lim_{x \rightarrow \infty} \left(\frac{x^3 + x^2 + x + \sin x}{x^2 + 2 \cos x} - a \sin x - bx + c \right) = 4$.

Now, match the following lists and then choose the correct code.

	List - I		List - II
(a)	The value of a is	p.	1
(b)	The value of b is	q.	3
(c)	The value of c is	r.	2
(d)	Number of real roots of equation $cx^2 + bx + a = 0$ is	s.	0

Codes:

	a	b	c	d
1	s	r	q	p
2	q	s	r	p
3	s	r	p	q
4	s	p	q	r

25.

	List - I		List - II
a.	$xy - \log y = 1$	p.	$xy_2 - \frac{1}{2}y_1 - \frac{1}{4}y = 0$
b.	$y = (\sin^{-1} x) / \sqrt{1-x^2}$	q.	$(1-x^2)y_1 - xy = 1$
c.	$y = \sqrt{2x-x^2}$	r.	$y^2 + (xy-1)y_1 = 0$
d.	$y = e^{\sqrt{x}} + e^{-\sqrt{x}}$	s.	$y^3y_2 + 1 = 0$

Where y_n is the n th derivative of y with respect to x .

26. Consider function $f(x) = \begin{cases} 2x+3, & x \leq 1 \\ -x^2+6, & x > 1 \end{cases}$. Now, match the following lists and then choose the correct code.

	List - I		List - II
a.	Number of points of non-differentiability of $f(x)$	p.	1
b.	Number of points of non-differentiability of $f(x)$	q.	3
c.	Number of points of non-differentiability of $ f(x) $	r.	4
d.	Number of points of non-differentiability of $ f(x) $	s.	5

Codes :

	a	b	c	d
(1)	p	q	q	r
(2)	p	s	q	r
(3)	p	q	q	s
(4)	p	r	q	r

27. Let $f_1: \mathbb{R} \rightarrow \mathbb{R}, f_2: [0, \infty) \rightarrow \mathbb{R}, f_3: \mathbb{R} \rightarrow \mathbb{R}$ and $f_4: \mathbb{R} \rightarrow [0, \infty)$ be defined by $f_1(x) = \begin{cases} |x| & \text{if } x < 0, \\ e^x & \text{if } x \geq 0; \end{cases}$

$$f_2(x) = x^2; f_3(x) = \begin{cases} \sin x & \text{if } x < 0 \\ x & \text{if } x \geq 0; \end{cases} \text{ and } f_4(x) = \begin{cases} f_2(f_1(x)) & \text{if } x < 0 \\ f_2(f_1(x)) - 1 & \text{if } x \geq 0 \end{cases}$$

	List - I		List - II
P.	f_4 is	1.	Onto but not one - one
Q.	f_3 is	2.	Neither continuous nor one - one
R.	$f_2 \circ f_1$ is	3.	Differentiable but not one - one
S.	f_2 is	4.	Continuous and one - one

(A) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 4; S \rightarrow 2$

(B) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 4; S \rightarrow 2$

(C) $P \rightarrow 3; Q \rightarrow 1; R \rightarrow 2; S \rightarrow 4$

(D) $P \rightarrow 1; Q \rightarrow 3; R \rightarrow 2; S \rightarrow 4$

28. Match the following:

	List – I		List – II
a.	At $x = 1, f(x) = \begin{cases} \log x, x < 1 \\ 2x - x^2, x \geq 1 \end{cases}$	p	is increasing
b.	At $x = 2, f(x) = \begin{cases} x - 1, x < 2 \\ 0, x = 2 \\ \sin x, x > 2 \end{cases}$	q	is decreasing
c.	At $x = 0, f(x) = \begin{cases} 2x + 3, x < 0 \\ 5, x = 0 \\ x^2 + 7, x > 0 \end{cases}$	r.	has point of maxima
d.	At $x = 0, f(x) = \begin{cases} e^{-x}, x < 0 \\ 0, x = 0 \\ -\cos x, x > 0 \end{cases}$	s.	Has point of minima

29. The value of $\lim_{n \rightarrow \infty} \frac{1}{1 + n \sin^2 nx}$ can be ($n \in \mathbb{N}$)

- (A) -1 (B) 0 (C) 1 (D)
- ∞

30. Given a real – valued function f such that

$$f(x) = \begin{cases} \frac{\tan^2\{x\}}{(x^2 - [x]^2)}, & \text{for } x > 0 \\ 1 & \text{for } x = 0 \\ \sqrt{\{x\} \cot\{x\}}, & \text{for } x < 0 \end{cases}$$

Where $[x]$ is the integral part and $\{x\}$ is the fractional part of x , then

- (A) $\lim_{x \rightarrow 0^+} f(x) = 1$ (B) $\lim_{x \rightarrow 0^-} f(x) = \cot 1$
 (C) $\cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = 1$ (D) $\tan^{-1} \left(\lim_{x \rightarrow 0^+} f(x) \right) = \frac{\pi}{4}$

31. Given $\lim_{x \rightarrow 0} \frac{f(x)}{x^2} = 2$, where $[.]$ denotes the greatest integer function, then

- (A) $\lim_{x \rightarrow 0} [f(x)] = 0$ (B) $\lim_{x \rightarrow 0} [f(x)] = 1$
 (C) $\lim_{x \rightarrow 0} \left[\frac{f(x)}{x} \right]$ does not exist (D) $\lim_{x \rightarrow 0} \left[\frac{f(x)}{x} \right]$ exists

32. Let $f(x) = \frac{1-x(1+|1-x|)}{|1-x|} \cos\left(\frac{1}{1-x}\right)$ for $x \neq 1$. Then
- (A) $\lim_{x \rightarrow 1^+} f(x) = 0$ (B) $\lim_{x \rightarrow 1^-} f(x)$ does not exist
 (C) $\lim_{x \rightarrow 1^-} f(x) = 0$ (D) $\lim_{x \rightarrow 1^+} f(x)$ does not exist
33. If $y = \cos^{-1}\left(\frac{2x}{1+x^2}\right)$, then $\frac{dy}{dx}$ is
- (A) $\frac{-2}{1+x^2}$ for all x (B) $\frac{-2}{1+x^2}$ for all $|x| < 1$
 (C) $\frac{2}{1+x^2}$ for $|x| > 1$ (D) None of these
34. $f_n(x) = e^{f_{n-1}(x)}$ for all $n \in \mathbb{N}$ and $f_0(x) = x$, then $\frac{d}{dx}\{f_n(x)\}$ is
- (A) $f_n(x) \frac{d}{dx}\{f_{n-1}(x)\}$ (B) $f_n(x)f_{n-1}(x)$
 (C) $f_n(x)f_{n-1}(x) \dots f_2(x)f_1(x)$ (D) None of these
35. Suppose f and g are functions having second derivatives f'' and g'' everywhere. If $f(x)g(x) = 1$ for all x and f' and g' are never zero, then $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)}$ equals
- (A) $\frac{-2f'(x)}{f(x)}$ (B) $-\frac{2g'(x)}{g(x)}$ (C) $\frac{-f'(x)}{f(x)}$ (D) $\frac{2f'(x)}{f(x)}$
36. For every pair of continuous functions $f, g: [0,1] \rightarrow \mathbb{R}$ such that $\max\{f(x) : x \in [0,1]\} = \max\{g(x) : x \in [0,1]\}$, the correct statement(s) is (are):
- (A) $(f(c))^2 + 3f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
 (B) $(f(c))^2 + f(c) = (g(c))^2 + 3g(c)$ for some $c \in [0,1]$
 (C) $(f(c))^2 + 3f(c) = (g(c))^2 + g(c)$ for some $c \in [0,1]$
 (D) $(f(c))^2 = (g(c))^2$ for some $c \in [0,1]$
37. For every integer n , let a_n and b_n be real numbers. Let functions $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by
- $$f(x) = \begin{cases} a_n + \sin \pi x, & \text{for } x \in [2n, 2n+1] \\ b_n + \cos \pi x, & \text{for } x \in (2n-1, 2n) \end{cases}$$
- For all integers n . If f is continuous, then which of the following hold(s) for all n ?
- (A) $a_{n-1} - b_{n-1} = 0$ (B) $a_n - b_n = 1$ (C) $a_n - b_{n+1} = 1$ (D) $a_{n-1} - b_n = -1$

38. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function with $g(0) = 0$,

$$g'(0) = 0 \text{ and } g'(1) \neq 0. \text{ Let } f(x) = \begin{cases} \frac{x}{|x|}g(x), & x \neq 0 \\ 0, & x = 0 \end{cases} \text{ and } h(x) = e^{|x|} \text{ for all } x \in \mathbb{R}. \text{ Let } (f \circ h)(x)$$

denote $f(h(x))$ and $(h \circ f)(x)$ denote $h(f(x))$. Then which of the following is (are) true?

- (A) f is differentiable at $x = 0$ (B) h is differentiable at $x = 0$
 (C) $f \circ h$ is differentiable at $x = 0$ (D) $h \circ f$ is differentiable at $x = 0$

39. Let $f: \mathbb{R} \rightarrow \mathbb{R}$, $g: \mathbb{R} \rightarrow \mathbb{R}$ and $h: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable functions such that

$$f(x) = x^3 + 3x + 2, g(f(x)) = x \text{ and } h(g(g(x))) = x \text{ for all } x \in \mathbb{R}. \text{ Then}$$

- (A) $g'(2) = \frac{1}{15}$ (B) $h'(1) = 666$ (C) $h(0) = 16$ (D) $h(g(3)) = 36$

40. A rectangular sheet of fixed perimeter with sides having their lengths in the ratio 8 : 15 is converted into an open rectangular box by folding after removing squares of equal area from all four corners. If the total area of removed squares is 100, the resulting box has maximum volume. Then the lengths of the sides of the rectangular sheet are

- (A) 24 (B) 32 (C) 45 (D) 60.

41. Let $f, g: [-1, 2] \rightarrow \mathbb{R}$ be continuous functions which are twice differentiable on the interval $(-1, 2)$. Let the values of f and g at the points $-1, 0$ and 2 be as given in the following table:

	$x = -1$	$x = 0$	$x = 2$
$f(x)$	3	6	0
$g(x)$	0	1	-1

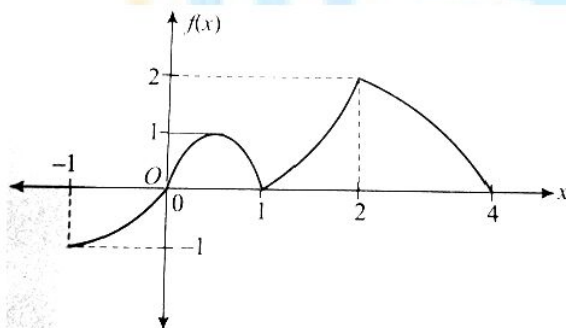
In each of the intervals $(-1, 0)$ and $(0, 2)$ the function $(f - 3g)''$ never vanishes. Then the correct statement(s) is (are)

- (A) $f'(x) - 3g'(x) = 0$ has exactly three solution in $(-1, 0) \cup (0, 2)$
 (B) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(-1, 0)$
 (C) $f'(x) - 3g'(x) = 0$ has exactly one solution in $(0, 2)$
 (D) $f'(x) - 3g'(x) = 0$ has exactly two solutions in $(-1, 0)$ and exactly two solutions in $(0, 2)$
42. Which of the following pair(s) of curves is /are orthogonal?

- (A) $y^2 = 4ax; y = e^{-x/2a}$ (B) $y^2 = 4ax; x^2 = 4ay$ at $(0, 0)$
 (C) $xy = a^2; x^2 - y^2 = b^2$ (D) $y = ax; x^2 + y^2 = c^2$

42. The angle between the tangents at any point P and the line joining P to the origin, where P is a point on the curve $\ln(x^2 + y^2) = c \tan^{-1} \frac{y}{x}$, c is constant, is
- (A) independent of x
 (B) independent of y
 (C) independent of x but dependent on y
 (D) independent of y but dependent on x
43. Which of the following is/are correct?
- (A) Between any two roots of $e^x \cos x = 1$, there exists at least one root of $\tan x = 1$
 (B) Between any two roots of $e^x \sin x = 1$, there exists at least one root of $\tan x = -1$
 (C) Between any two roots of $e^x \cos x = 1$, there exists at least one root of $e^x \sin x = 1$.
 (D) Between any two roots of $e^x \sin x = 1$, there exists at least one root of $e^x \cos x = 1$.
44. Let $f(x) = \begin{cases} x^2 + 3x, & -1 \leq x < 0 \\ -\sin x, & 0 \leq x < \pi/2 \\ -1 - \cos x, & \frac{\pi}{2} \leq x \leq \pi \end{cases}$. Then
- (A) f(x) has global minimum value -2
 (B) global maximum value occurs at $x = 0$
 (C) global maximum value occurs at $x = \pi$
 (D) $x = \pi/2$ is point of local minima
45. If $f(x) = 2x + \cot^{-1} x + \log(\sqrt{1+x^2} - x)$, then f(x)
- (A) Increases in $[0, \infty)$ (B) Decreases in $[0, \infty)$
 (C) Neither increases nor decreases in $[0, \infty)$ (D) Increases in $(-\infty, \infty)$
46. Let f(x) be a non-negative continuous and bounded function for all $x \geq 0$. If $(\cos x)f'(x) \leq (\sin x - \cos x)f(x) \forall x \geq 0$, then which of the following is / are correct?
- (A) $f(6) + f(5) > 0$
 (B) $x^2 - 3x + 2 + f(7) = 0$ has 2 distinct solutions
 (C) $f(5)f(7) - f(6)f(5) = 0$
 (D) $\lim_{x \rightarrow 6} \frac{f(x) - \sin(\pi x)}{(x - 6)} = 1$

47. If $f(x) = \frac{x^2}{2-2\cos x}$ and $g(x) = \frac{x^2}{6x-6\sin x}$, where $0 < x < 1$, then
- (A) f is increasing function
 (B) g is increasing function
 (C) f is decreasing function
 (D) g is decreasing function
48. Consider the function $f(x) = x \cos x - \sin x$, then identify the statement which is/are correct.
- (A) f is odd
 (B) f is monotonic decreasing at $x = 0$
 (C) f has point of inflection at $x = 0$
 (D) f has a maxima at $x = \pi$
49. If $f(x) = \lim_{t \rightarrow 0} \frac{|a + \sin \pi x|^t - 1}{|a + \sin \pi x|^t + 1}$, $x \in (0, 6)$, then
- (A) if $a = 1$, then $f(x)$ has 5 points of discontinuity
 (B) if $a = 3$, then $f(x)$ has not point of discontinuity
 (C) if $a = 0.5$ then $f(x)$ has 6 points of discontinuity
 (D) if $a = 0$, then $f(x)$ has 6 points of discontinuity
50. If graph of a function $f(x)$ which is defined in $[-1, 4]$ is shown in the following figure then identify the correct statement(s)



- (A) domain of $f(|x|-1)$ is $[-5, 5]$
 (B) range of $f(|x|+1)$ is $[0, 2]$
 (C) range of $f(-|x|)$ is $[-1, 0]$
 (D) domain of $f(|x|)$ is $[-3, 3]$
51. Let $f(x) = \frac{3}{4}x + 1$, $f^n(x)$ be defined as $f^2(x) = f(f(x))$, and for $n \geq 2$, $f^{n+1}(x) = f(f^n(x))$. If $\lambda = \lim_{n \rightarrow \infty} f^n(x)$, then
- (A) λ is independent of x

- (B) λ is a linear polynomial in x
 (C) the line $y = \lambda$ has slope 0
 (D) the line $4y = \lambda$ touches the unit circle with center at the origin.

52. If $f(\alpha) = \lim_{x \rightarrow 2} (\sin^x \alpha + \cos^x \alpha)^{\frac{1}{(x-2)}}$ for $\alpha \in \left[0, \frac{\pi}{2}\right]$, then

(A) $f(0) = 1$

(B) $f\left(\frac{\pi}{2}\right) = 1$

(C) $f(\alpha) = (\cos \alpha)^{\cos^2 \alpha} \cdot (\sin \alpha)^{\sin^2 \alpha}$ if $\alpha \in \left(0, \frac{\pi}{2}\right)$

(D) $f(\alpha) = \frac{(\sin \alpha)^{\sin^2 \alpha}}{(\cos \alpha)^{\cos^2 \alpha}}$ if $\alpha \in \left(0, \frac{\pi}{2}\right)$

53. $\lim_{x \rightarrow 1} \frac{(1-x)(1-x^2) \dots (1-x^{2n})}{\{(1-x)(1-x^2) \dots (1-x^n)\}^2}$, $n \in \mathbb{N}$ equals

(A) ${}^{2n}P_n$

(B) ${}^{2n}C_n$

(C) $(2n)!$

(D) None of these

54. The value of $\lim_{x \rightarrow 0} \frac{1 - (\cos x)\sqrt{\cos 2x}}{x^2}$ is

(A) $-\frac{3}{2}$

(B) $-\frac{1}{2}$

(C) $\frac{1}{2}$

(D) $\frac{3}{2}$

55. $\lim_{x \rightarrow 0} \frac{\cos(\tan x) - \cos x}{x^4}$ is equal to

(A) $1/6$

(B) $-1/3$

(C) $1/2$

(D) 1

56. For each $t \in \mathbb{R}$, let $[t]$ be the greatest integer less than or equal to t . Then

$\lim_{x \rightarrow 0^+} x \left(\left[\frac{1}{x} \right] + \left[\frac{2}{x} \right] + \dots + \left[\frac{15}{x} \right] \right)$

(A) Is equal to 15

(B) Is equal to 120

(C) does not exist (in $\mathbb{R}$)

(D) is equal to 0

57. $\lim_{x \rightarrow 2} \left(\sum_{n=1}^9 \frac{x}{n(n+1)x^2 + 2(2n+1)x + 4} \right)$ is equal to:

(A) $\frac{9}{44}$

(B) $\frac{5}{24}$

(C) $\frac{1}{5}$

(D) $\frac{7}{36}$

58. If $f(x) = x^3 + 3x + 4$ and g is the inverse function of $f(x)$, then the value of $\frac{d}{dx} \left(\frac{g(x)}{g(g(x))} \right)$ at $x =$

4 equals

- (A) $-\frac{1}{6}$ (B) 6 (C) $-\frac{1}{3}$ (D) non – existenet

59. $\frac{d^{20}y}{dx^{20}}(2 \cos x \cos 3x)$ is equal to

- (A) $2^{20}(\cos 2x - 2^{20} \cos 3x)$ (B) $2^{20}(\cos 2x + 2^{20} \cos 4x)$
(C) $2^{20}(\sin 2x + 2^{20} \sin 4x)$ (D) $2^{20}(\sin 2x - 2^{20} \sin 4x)$

60. If $y^{1/m} = (x + \sqrt{1+x^2})$, then $(1+x^2)y_2 + xy_1$ is (where y_r represents the r th derivative of y w.r.t. x)

- (A) m^2y (B) my^2 (C) m^2y^2 (D) None of these

61. Let $\alpha \in \mathbb{R}$ be such that the function

$$f(x) = \begin{cases} \frac{\cos^{-1}(1-\{x\})\sin^{-1}(1-\{x\})}{\{x\}-\{x\}^3}, & x \neq 0 \\ \alpha, & x = 0 \end{cases} \text{ is}$$

Continuous at $x = 0$, where $\{x\} = x - [x]$, $[x]$ is the greatest integer less than or equal to x .

- (A) No such α exists (B) $\alpha = \frac{\pi}{\sqrt{2}}$
(C) $\alpha = 0$ (D) $\alpha = \frac{\pi}{4}$

62. Let $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$ and

$g(x) = |f(x)| + f(|x|)$. Then, in the interval $(-2, 2)$, g is:

- (A) Differentiable at all points (B) not continuous
(C) not differentiable at two points (D) not differentiable at one point

63. The derivative of $\tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$ with respect to $\tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$ at $x = \frac{1}{2}$ is:

- (A) $\frac{2\sqrt{3}}{5}$ (B) $\frac{\sqrt{3}}{12}$ (C) $\frac{2\sqrt{3}}{3}$ (D) $\frac{\sqrt{3}}{10}$

64. If $\left[x + \sqrt{x^2 - 1}\right]^{15} + \left[x - \sqrt{x^2 - 1}\right]^{15}$, then $(x^2 - 1)\frac{d^2y}{dx^2} + x\frac{dy}{dx}$ is equal to
 (A) $12y$ (B) $224y^2$ (C) $225y^2$ (D) $225y$
65. If $x = \phi(t)$, $y = \psi(t)$, then $\frac{d^2y}{dx^2}$ is
 (A) $\frac{\phi'\psi'' - \psi'\phi''}{(\phi')^2}$ (B) $\frac{\phi'\psi'' - \psi'\phi''}{(\phi')^3}$
 (C) $\frac{\phi''}{\psi''}$ (D) $\frac{\psi''}{\phi''}$
66. Given that $\prod_{n=1}^n \cos \frac{x}{2^n} = \frac{\sin x}{2^n \sin\left(\frac{x}{2^n}\right)}$ and $f(x) = \begin{cases} \lim_{n \rightarrow \infty} \sum_{n=1}^n \frac{1}{2^n} \tan\left(\frac{x}{2^n}\right), & x \in (0, \pi) - \left\{\frac{\pi}{2}\right\} \\ \frac{2}{\pi}, & x = \frac{\pi}{2} \end{cases}$
 Then which one of the following is true?
 (A) $f(x)$ has non – removable discontinuity of finite type at $x = \frac{\pi}{2}$
 (B) $f(x)$ has removable discontinuity at $x = \frac{\pi}{2}$
 (C) $f(x)$ is continuous at $x = \frac{\pi}{2}$
 (D) $f(x)$ has non – removable discontinuity of infinite type at $x = \frac{\pi}{2}$.
67. If number of points of discontinuity of the function $f(x) = [2 + 10 \sin x]$, in $x \in \left[0, \frac{\pi}{2}\right]$ is same as number of points of non – differentiability of the function $g(x) = (x-1) \times (x-2) | (x-1)(x-2) \dots (x-2m) |$, ($m \in \mathbb{N}$) in $x \in (-\infty, \infty)$, then the value of m is
 (A) 4 (B) 5 (C) 6 (D) 7
68. A lamp of negligible height is placed on the ground ℓ_1 m away from a wall. A man ℓ_2 m tall is walking at a speed of $\frac{\ell_1}{10}$ m/s from the lamp to the nearest point on the wall. When he is midway between the lamp and the wall, the rate of change in the in the length of this shadow on the wall is
 (A) $-\frac{5\ell_2}{2}$ m/s (B) $-\frac{2\ell_2}{5}$ m/s (C) $-\frac{\ell_2}{2}$ m/s (D) $-\frac{\ell_2}{5}$ m/s

69. A factory D is to be connected by a road with a straight railway line on which a town A is situated. The distance DB of the factory to the railway line is $5\sqrt{3}$ km. Length AB of the railway line is 20 km. Freight charges on the road are twice the charges on the railway. The point P ($AP < AB$) on the railway line should the road DP be connected so as to ensure minimum freight charges from the factory to the town is
 (A) $BP = 5$ km (B) $AP = 5$ km (C) $BP = 7.5$ km (D) None of these
70. The largest area of a trapezium inscribed in a semi-circle of radius R, if the lower base is the diameter, is
 (A) $\frac{3\sqrt{3}}{4}R^2$ (B) $\frac{\sqrt{3}}{2}R^2$ (C) $\frac{3\sqrt{3}}{8}R^2$ (D) R^2
71. If $\phi(x)$ is a polynomial function and $\phi(x) < \phi'(x) \forall x \leq 1$ and $\phi(1) = 0$, then
 (A) $\phi(x) \geq 0 \forall x \geq 1$ (B) $\phi(x) < 0 \forall x \geq 1$ (C) $\phi(x) = 0 \forall x \geq 1$ (D) None of these
72. Suppose that f is a polynomial of degree 3 and that $f''(x) \neq 0$ at any of the stationary point. Then
 (A) f has exactly one stationary point
 (B) f must have no stationary point
 (C) f must have exactly two stationary points
 (D) f has either zero or two stationary points
73. If $P(x)$ is a polynomial of degree less than or equal to 2 and S is the set of all such polynomials so that $P(0) = 0$, $P(1) = 1$ and $P'(x) > 0 \forall x \in [0, 1]$, then
 (A) $S = \phi$
 (B) $S = ax + (1-a)x^2 \forall a \in (0, 2)$
 (C) $S = ax + (1-a)x^2 \forall a \in (0, \infty)$
 (D) $S = ax + (1-a)x^2 \forall a \in (0, 1)$
74. The total number of local maxima and local minima of the function

$$f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$$
 is
 (A) 0 (B) 1 (C) 2 (D) 3

75. Let $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in \mathbb{R}$. Then the set of all x satisfying $(f \circ g \circ g \circ f)(x) = (g \circ g \circ f)(x)$, where $(f \circ g)(x) = f(g(x))$, is
- (A) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, \dots\}$ (B) $\pm\sqrt{n\pi}, n \in \{1, 2, \dots\}$
- (C) $\frac{\pi}{2} + 2n\pi, n \in \{\dots - 2, -1, 0, 1, 2, \dots\}$ (D) $2n\pi, n \in \{\dots - 2, -1, 0, 1, 2, \dots\}$
76. X and Y are two sets and $f: X \rightarrow Y$. If $\{f(c) = y; c \in X, y \in Y\}$ and $\{f^{-1}(d) = x; d \in Y, x \in X\}$, then the true statement is
- (A) $f(f^{-1}(b)) = b$ (B) $f^{-1}(f(a)) = a$
- (C) $f(f^{-1}(b)) = b, b \in Y$ (D) $f^{-1}(f(a)) = a, a \in X$
77. Consider function $f: A \rightarrow B$ and $g: B \rightarrow C$ such that $(g \circ f)^{-1}$ exists, then:
- (A) f and g both are one – one
- (B) f and g both are onto
- (C) f is one – one and g is onto
- (D) f is onto and g is one – one
- P-1: $A_i = \frac{x - a_i}{|x - a_i|}, i = 1, 2, \dots, n$, and $a_1 < a_2 < a_3 < \dots < a_n$.
78. If $1 \leq m \leq n, m \in \mathbb{N}$, then the value of $L = \lim_{x \rightarrow a_m} (A_1 A_2 \dots A_n)$ is
- (A) always 1 (B) always -1 (C) $(-1)^{n-m+1}$ (D) $(-1)^{n-m}$
79. If $1 \leq m \leq n, m \in \mathbb{N}$, then the value of $R = \lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n)$ is
- (A) always 1 (B) always -1 (C) $(-1)^{m+1}$ (D) $(-1)^{n-m}$
80. If $1 \leq m \leq n, m \in \mathbb{N}$, then $\lim_{x \rightarrow a_m} (A_1 A_2 \dots A_n)$
- (A) is always equal to -1 (B) is always equal to +1
- (C) does not exist (D) None of these
- P-2: If $L = \lim_{x \rightarrow 0} \frac{\sin x + ae^x + be^{-x} + c \log_e(1+x)}{x^3}$ exists finitely, then
81. The value of L is
- (A) $1/2$ (B) $-1/3$ (C) $-1/6$ (D) 3

82. Equation $ax^2 + bx + c = 0$ has

- (A) real and equal roots (B) complex roots
(C) unequal positive real roots (D) unequal roots

83. The solutions set of $||x + c| - 2a| < 4b$ is

- (A) $[-2, 2]$ (B) $[0, 2]$ (C) $[-1, 1]$ (D) $[-2, 1]$

P-3: Consider $f(x) = x^2 + ax + 3$ and $g(x) = x + b$ and $F(x) = \lim_{n \rightarrow \infty} \frac{f(x) + x^{2n}g(x)}{1 + x^{2n}}$.

84. If $F(x)$ is continuous at $x = 1$, then

- (A) $b = a + 3$ (B) $b = a - 1$ (C) $a = b - 2$ (D) None of these

85. If $F(x)$ is continuous at $x = -1$, then

- (A) $a + b = -2$ (B) $a - b = 3$ (C) $a + b = 5$ (D) None of these

86. If $F(x)$ is continuous at $x = \pm 1$, then $f(x) = 2g(x)$ has

- (A) imaginary roots (B) both the roots positive
(C) both the roots negative (D) roots of opposite signs

P-4: Consider, $f(x) = \frac{(x^2 - 1)(x^2 - 4)}{g(x)}$, $x \in \mathbb{R} - \{a, b\}$, where $g(x)$ is a polynomial of degree ≤ 4 .

$f(x)$ has a removable type of discontinuity at $x = a$ and $x = b$. Also it is known that

(i) $\lim_{x \rightarrow \pm\infty} f(x) = +\infty$

(ii) $\lim_{x \rightarrow -1} f(x) = 6$

(iii) $\lim_{x \rightarrow -2} f(x) = \text{a non-zero finite number}$

Let L and M denote the number of points of discontinuity and the number of points of non-differentiability respectively of any function in $\mathbb{R}$. ($\mathbb{R}$ is the set of real numbers)

87. Number of points where $|f(x)|$ is non-differentiable in its domain is

- (A) 2 (B) 3 (C) 4 (D) 5

88. Number of points where $f(|x|)$ is non-differentiable in its domain is

- (A) 2 (B) 3 (C) 4 (D) 5

89. Number of points where $f(-|x|)$ is non-differentiable in its domain is

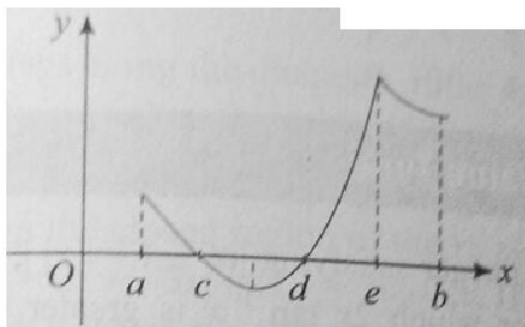
- (A) 0 (B) 1 (C) 2 (D) 3

90. A curve is represented parametrically by the equations

$x = f(t) = a^{\ln(b^t)}$ and $y = g(t) = b^{-\ln(a^t)}$, $b > 0$ and $a \neq 1, b \neq 1$ where $t \in \mathbb{R}$.

90. Which of the following is not a correct expression for $\frac{dy}{dx}$?
- (A) $\frac{-1}{f(t)^2}$ (B) $-(g(t))^2$ (C) $\frac{-g(t)}{f(t)}$ (D) $\frac{-f(t)}{g(t)}$
91. The value of $\frac{d^2y}{dx^2}$ at the point where $f(t) = g(t)$ is
- (A) 0 (B) $\frac{1}{2}$ (C) 1 (D) 2
92. The value of $\frac{f(t)}{f'(t)} \cdot \frac{f''(-t)}{f'(-t)} + \frac{f(-t)}{f'(-t)} \cdot \frac{f''(t)}{f'(t)} \forall t \in \mathbb{R}$ is equal to
- (A) -2 (B) 2 (C) -4 (D) 4
93. If a continuous function f defined on the real line $\mathbb{R}$, assumes positive and negative values in $\mathbb{R}$ then the equation $f(x) = 0$ has a root in $\mathbb{R}$. For example, if it is known that a continuous function f on $\mathbb{R}$ is positive at some point and its minimum value is negative then the equation $f(x) = 0$ has a root in $\mathbb{R}$.
- Consider $f(x) = ke^x - x$ for all real x where k is a real constant.
93. The line $y = x$ meets $y = ke^x$ for $k \leq 0$ at
- (A) no point (B) one point (C) two points (D) more than two points
94. The positive value of k for which $ke^x - x = 0$ has only one root is
- (A) $\frac{1}{e}$ (B) 1 (C) e (D) $\log_e 2$
95. For $k > 0$, the set of all values of k for which $ke^x - x = 0$ has two distinct root is
- (A) $\left(0, \frac{1}{e}\right)$ (B) $\left(\frac{1}{e}, 1\right)$ (C) $\left(\frac{1}{e}, \infty\right)$ (D) $(0, 1)$
96. Consider the function $f(x) = x + \cos x - a$
- Which of the following is not true about $y = f(x)$?
- (A) It is an increasing function
- (B) It is a monotonic function
- (C) It has infinite points of inflections
- (D) None of these.

97. Value of a for which $f(x) = 0$ has exactly one positive root are
 (A) $(0,1)$ (B) $(-\infty,1)$ (C) $(-1,1)$ (D) $(1,\infty)$
98. Values of a for which $f(x) = 0$ has exactly one negative root are
 (A) $(0,1)$ (B) $(-\infty,1)$ (C) $(-1,1)$ (D) $(1,\infty)$
99. The graph of $y = g(x) = f'(x)$ is as shown in the following figure. Analyse this graph and answer the following questions



99. The graph of $y = f(x)$ for $a < x < b$ has
 (A) no point of extrema (B) one point of extrema
 (C) two points extrema (D) can't say anything
100. number of points of inflection the graph of $y = f(x)$ for $a < x < b$ has
 (A) 0 (B) 1
 (C) 2 (D) Can't say anything

KEY & SOLUTIONS

1. 25

Sol: We have $\lim_{x \rightarrow 1} \frac{100}{(1-x^{50})(1+x^{50})} - \frac{50}{1-x^{50}}$

$$= \lim_{x \rightarrow 1} \frac{100 - 50(1+x^{50})}{(1-x^{50})(1+x^{50})}$$

$$= \lim_{x \rightarrow 1} \frac{50(1-x^{50})}{(1-x^{50})(1+x^{50})}$$

$$= \lim_{x \rightarrow 1} \frac{50}{1+x^{100}}$$

2. 3

Sol: $S_n = \frac{n(n+1)}{2}$ and $S_{n-1} = \frac{(n+2)(n-1)}{2}$

$$\therefore \frac{S_n}{S_{n-1}} = \frac{n(n+1)}{2} \cdot \frac{2}{(n+2)(n-1)} = \left(\frac{n}{n-1}\right)\left(\frac{n+1}{n+2}\right)$$

$$\therefore P_n = \left(\frac{2}{1} \cdot \frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdots \frac{n}{n-1}\right) \left(\frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{6} \cdots \frac{n+1}{n+2}\right)$$

$$= \left(\frac{n}{1}\right) \left(\frac{3}{n+2}\right)$$

3. 0

Sol: $\lim_{n \rightarrow \infty} \left[\sqrt[3]{(n+1)^2} - \sqrt[3]{(n-1)^2} \right]$

$$= \lim_{n \rightarrow \infty} n^{2/3} \left[\left(1 + \frac{1}{n}\right)^{2/3} - \left(1 - \frac{1}{n}\right)^{2/3} \right]$$

$$= \lim_{n \rightarrow \infty} n^{2/3} \left[\left(1 + \frac{2}{3} \cdot \frac{1}{n} + \frac{\frac{2}{3} \left(\frac{2}{3} - 1\right)}{2!} \cdot \frac{1}{n^2} \cdots \right) - \left(1 - \frac{2}{3} \cdot \frac{1}{n} + \frac{\frac{2}{3} \left(\frac{2}{3} - 1\right)}{2!} \cdot \frac{1}{n^2} \cdots \right) \right]$$

$$= \lim_{n \rightarrow \infty} n^{2/3} \left[\frac{4}{3} \cdot \frac{1}{n} + \frac{8}{81} \cdot \frac{1}{n^3} + \cdots \right]$$

4. 2

Sol: $L = \lim_{n \rightarrow \infty} \prod_{n=2}^n \frac{n^2-1}{n^2}$

$$= \lim_{n \rightarrow \infty} \prod_{n=2}^n \frac{n-1}{n} \cdot \prod_{n=2}^n \frac{n+1}{n}$$

$$= \lim_{n \rightarrow \infty} \left(\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdots \frac{n-1}{n} \right) \left(\frac{3}{2} \cdot \frac{4}{3} \cdot \frac{5}{4} \cdots \frac{n+1}{n} \right)$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \frac{n+1}{2} = \frac{1}{2}$$

5. 0

Sol: $\lim_{x \rightarrow 1} \left\{ \frac{-ax + \sin(x-1) + a}{x + \sin(x-1) - 1} \right\}^{\frac{1-x}{1-\sqrt{x}}} = \frac{1}{4}$

$$\Rightarrow \lim_{x \rightarrow 1} \left\{ \frac{\frac{\sin(x-1)}{(x-1)} - a}{\frac{\sin(x-1)}{(x-1)} + 1} \right\}^{1+\sqrt{x}} = \frac{1}{4}$$

$$\Rightarrow \left(\frac{1-a}{2} \right)^2 = \frac{1}{4}$$

$$\Rightarrow a = 0, a = 2$$

6. 18

Sol: Let degree of $f(X)$ is n , degree of $f'(x)$ is $n-1$, and degree of $f''(x)$ is $(n-2)$. Hence,

$$n = (n-1) + (n-2) = 2n-3$$

$$\therefore n = 3$$

Hence, $f(x) = ax^3 + bx^2 + cx + d$ ($a \neq 0$)

$$f'(x) = 3ax^2 + 2bx + c$$

$$f''(x) = 6ax + 2b$$

$$\therefore ax^3 + bx^2 + cx + d = (3ax^2 + 2bx + c)(6ax + 2b)$$

$$\therefore 18a^2 = a \text{ or } a = \frac{1}{18}$$

7. 8

Sol: $(y - x^5)^2 = x(1 + x^2)^2$

Differentiating both sides w.r.t. x , we get

$$2(y - x^5) \left(\frac{dy}{dx} - 5x^4 \right) = 1(1 + x^2) + (x)(2(1 + x^2))(2x)$$

On putting $x = 1$, $y = 3$ in above equation, we get

$$\frac{dy}{dx} = 8$$

8. 50

$$f(x) = x + \frac{1}{x + f(x)} = x + \frac{1}{x + f(x)}$$

Sol: $x + x + \frac{1}{2x + \frac{1}{2x + \dots \infty}}$

$$\text{Or } f(x) - x = \frac{1}{x + f(x)}$$

$$\text{Or } f^2(x) - x^2 = 1$$

Differentiating w.r.t. x , we get

$$\text{Or } 2f(x) \cdot f'(x) - 2x = 0$$

$$\text{Or } f(x) \cdot f'(x) = x$$

$$\text{Or } f(50) \cdot f'(50) = 50$$

9. 6

$$\text{Sol: } g(f(x)) = \begin{cases} g\left(\frac{x}{2} - 1\right), & 0 \leq x < 1 \\ g\left(\frac{1}{2}\right), & 1 \leq x \leq 2 \end{cases}$$

$$= \begin{cases} \frac{(x-1)(x-2-2k)}{2} + 3, & 0 \leq x < 1 \\ 4 - 2k, & 1 \leq x < 2 \end{cases}$$

$$\lim_{x \rightarrow 1^-} g(f(x)) = 3, g(f(1)) = 4 - 2k$$

$$\text{And } \lim_{x \rightarrow 1^+} g(f(x)) = 4 - 2k$$

For $g(f(x))$ to be continuous at $x = 1$, $4 - 2k = 3$ or $k = \frac{1}{2}$

10. 8

$$\text{Sol: } f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x) + f(h) + 2xh(x+h) - \frac{1}{3} - \left(f(x) + f(0) - \frac{1}{3}\right)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} + 2x^2$$

$$= f'(0) + 2x^2$$

$$\lim_{h \rightarrow 0} \frac{3f(h) - 1}{6h} = \lim_{h \rightarrow 0} \frac{f(h) - \frac{1}{3}}{2h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{2h}$$

$$= \frac{f'(0)}{2} = \frac{2}{3}$$

$$\therefore f'(0) = \frac{4}{3}$$

$$\therefore f'(x) = \frac{4}{3} + 2x^2$$

$$f(x) = \lambda + \frac{4}{3}x + \frac{2x^3}{3} \quad \text{or } f(0) = \lambda = \frac{1}{3}$$

$$\therefore f(x) = \frac{2x^3}{3} + \frac{4}{3}x + \frac{1}{3} \quad \text{or } f(2) = \frac{25}{3}$$

11. 5

$$\text{Sol: } f''(x) = \begin{cases} -x^{p-4} \sin\left(\frac{1}{x}\right) - (p-2)x^{p-3} \cos\left(\frac{1}{x}\right) - px^{p-3} \cos\left(\frac{1}{x}\right) \\ \quad + p(p-1)x^{p-2} \sin\left(\frac{1}{x}\right) + 2, & x > 0 \\ -x^{p-4} \sin\left(\frac{1}{x}\right) - (p-2)x^{p-3} \cos\left(\frac{1}{x}\right) \\ \quad + px^{p-3} \cos\left(\frac{1}{x}\right) \\ \quad + p(p-1)x^{p-2} \sin\left(\frac{1}{x}\right) - 2, & x < 0 \\ 0 & x = 0 \end{cases}$$

$$\text{RHL} = \text{LHL} = f'(0) = 0$$

Since $\sin \infty$ and $\cos \infty$ lie between -1 to 1 , for $p \geq 5$, $\text{RHL} = 2$

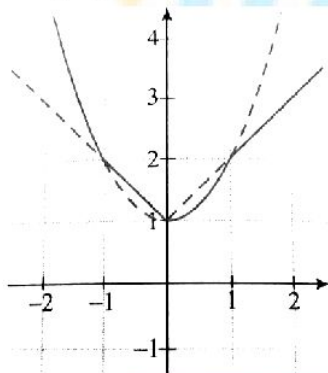
$$\text{LHL} = -2.$$

$$f'(0) = 0$$

For $p \in [5, \infty)$, $f''(x)$ is not continuous.

12. 3

Sol: The graph of $f(x) = |x| + 1$ and $g(x) = x^2 + 1$ are as shown in the following figure.



From the graph

$$h(x) = \begin{cases} \max\{f(x), g(x)\}, & \text{if } x \leq 0 \\ \min\{f(x), g(x)\}, & \text{if } x > 0 \end{cases}$$

$$= \begin{cases} x^2 + 1, & x \in (-\infty, -1] \\ -x + 1, & x \in (-1, 0] \\ x^2 + 1, & x \in (0, 1] \\ x + 1, & x \in (1, \infty) \end{cases}$$

Hence, $h(x)$ is not differentiable at $x = -1, 0, 1$

13. 11

$$\text{Sol: } g(x) = x^3 + 3x^2 + 7x + 2$$

$$g'(x) = 3x^2 + 6x + 7 > 0 \text{ for all real } x \text{ as its discriminant is negative.}$$

Thus, $g(x)$ is an increasing function.

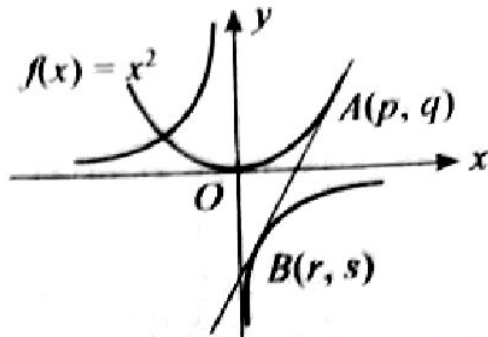
$$g(0) = 2 \text{ and } g(1) = 1 + 3 + 7 + 2 = 13$$

Thus, $f(x) = [g(x)]$ is discontinuous when $g(x) = 3, 4, \dots, 13$.

So there are 11 points of continuity

14. 5

Sol: or $pr^2 = 4$



$$\text{Now, } m_{AB} = \frac{q-s}{p-r}$$

$$\text{Or } 2p = \frac{p^2 + \frac{8}{r}}{p-r}$$

$$\text{Or } p^2 = 2pr + \frac{8}{r} \text{ or } p^2 = \frac{16}{r}$$

$$\text{Or } \frac{16}{r^4} = \frac{16}{r} \text{ or } r = 1 (r \neq 0) \text{ or } p = 4$$

$$\therefore r = 1, p = 4$$

$$\text{Hence, } p+r = 5$$

15. 7

Sol: $x = r^2, y = t^3$

$$\frac{dx}{dt} = 2t; \frac{dy}{dt} = 3t^2$$

$$\frac{dy}{dx} = \frac{3t}{2}$$

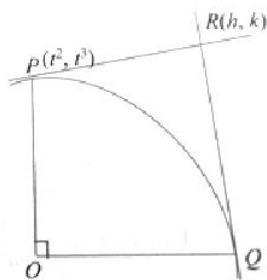
Equation of tangent at P is

$$y - t^3 = \frac{3t}{2}(x - r^2)$$

$$2k - 2t^3 = 3th - 3t^3$$

$$\therefore t^3 - 3th + 2k = 0$$

Product of roots, $t_1 t_2 t_3 = -2k$



Putting $t_1 t_2 = -1, t_3 = 2k$

Now, t_3 must satisfy equation (1). Therefore,

$$(2k)^3 - 3(2k)h + 2k = 0$$

$$\text{i.e. } 4y^2 - 3x + 1 = 0 \quad \text{or } 4y^2 = 3x - 1$$

$$\text{or } a + b = 7$$

16. 12

Sol: On differentiating (1) w.r.t. x , we get

$$f'(x) = -f'(6-x)$$

Putting $x = 0, 2, 3, 5$ in (2), we get

$$f'(0) = -f'(6) = 0$$

$$\text{Similarly, } f'(2) = f'(4) = 0$$

$$f'(3) = 0$$

$$f'(5) = f'(1) = 0$$

$$\therefore f'(0) = 0 = f'(2) = f'(3) = f'(5) = f'(1) = f'(4) = f'(6)$$

Therefore, $f'(x) = 0$ has minimum seven roots in $[0, 6]$

Now, consider a function $y = f'(x)$.

As $f'(x)$ satisfies Rolle's theorem in intervals $[0, 1], [1, 2], [2, 3], [3, 4], [4, 5]$, and $[5, 6]$, respectively, by Rolle's theorem, the equation $f''(x) = 0$ has minimum six roots.

Now, $g(x) = (f''(x))^2 + f'(x)f'''(x) = 0 = h'(x)$, where

$$h(x) = f'(x)f''(x).$$

Clearly, $h(x) = 0$ has minimum 13 roots in $[0, 6]$.

17. 2

$$\text{Sol: } f(x) = 30 - 2x - x^3$$

$$f'(x) = -2 - 3x^2 < 0$$

$\Rightarrow f(x)$ is decreasing function

$$\text{Hence } f(f(f(x))) > f(f(-x))$$

$$\Rightarrow f(f(x)) < f(-x)$$

$$\Rightarrow f(x) > -x$$

$$\Rightarrow 30 - 2x - x^3 > -x$$

$$\Rightarrow x^3 + x - 30 < 0$$

$$\Rightarrow (x-3)(x^2+3x+10) < 0$$

$$\Rightarrow x < 3$$

18. 8

Sol: $f(x) = \frac{1}{1+|x|} + \frac{1}{1+|x-1|}$

Clearly $f(x)$ is non-differentiable at $x = 0, 1$

Also $f(1/2+x) = f(1/2-x)$

Hence graph of $f(x)$ is symmetrical about the line $x = 1/2$, where derivative is zero.

Local minimum = $f\left(\frac{1}{2}\right) = \frac{4}{3}$

Global maximum = $f(0) = f(1) = \frac{3}{2}$

$$\therefore k = \frac{4}{3} + \frac{3}{2} = \frac{17}{6}$$

19. 4

Sol: Let the inner radius and inner length be r and h , respectively.

$$\therefore V = \pi r^2 h$$

Let the volume of material be M .

$$\Rightarrow M = \pi(r+2)^2 2 + \pi(r+2)^2 h - \pi r^2 h$$

$$= 2\pi(r+2)^2 + 4\pi h(r+1)$$

$$= 2\pi \left((r+2)^2 + \frac{2(r+1)V}{\pi r^2} \right)$$

$$\Rightarrow \frac{dM}{dr} = 2\pi \left(2(r+2) + \frac{2V}{\pi} \left(\frac{-1}{r^2} - \frac{2}{r^3} \right) \right)$$

Given volume is maximum when $r = 10$

$$\therefore \frac{dM}{dr} = 0, \text{ when } r = 10$$

$$\Rightarrow 24 + \frac{2V}{\pi} \left(\frac{-10-2}{10^3} \right) = 0$$

$$\Rightarrow \frac{24V}{10^3 \pi} = 24$$

$$\Rightarrow V = 10^3 \pi$$

$$\Rightarrow \frac{V}{250\pi} = 4$$

20. 5

Sol: $f(x) = |x| + |x^2 - 1|$

$$f'(x) = \frac{|x|}{x} + \frac{|x^2-1|}{x^2-1} \cdot (2x) = \begin{cases} 2x-1, & x < -1 \\ -(2x+1), & -1 < x < 0 \\ 1-2x, & 0 < x < 1 \\ 2x+1, & x > 1 \end{cases}$$

So, $f'(x)$ changes sign at points $x = -1, -1/2, 0, 1/2, 1$

So, total number of points of local maximum or minimum is 5

21. 5

Sol: Since $f(|x|) = 0$ has 8 roots

$\Rightarrow f(x) = 0$ has 4 real positive roots but since $f(x)$ is polynomial of degree 5

$\Rightarrow$ fifth root is also real.

$\therefore$ Number of real roots of $f(x) = 0$ is 5.

22. 3

Sol: $f(x) = \sqrt{\sin x + \cos x} + \sqrt{7x - x^2 - 6}$

$$= \sqrt{\sqrt{2} \sin\left(x + \frac{\pi}{4}\right)} + \sqrt{(x-6)(1-x)}$$

Now, $f(x)$ is defined if $\sin\left(x + \frac{\pi}{4}\right) \geq 0$ and $(x-6)(1-x) \geq 0$

i.e., $0 \leq x + \frac{\pi}{4} \leq \pi$ or $2\pi \leq x + \frac{\pi}{4} \leq 3\pi$ and $1 \leq x \leq 6$

i.e., $-\frac{\pi}{4} \leq x \leq \frac{3\pi}{4}$ or $\frac{7\pi}{4} \leq x \leq \frac{11\pi}{4}$ and $1 \leq x \leq 6$

or $x \in \left[1, \frac{3\pi}{4}\right] \cup \left[\frac{7\pi}{4}, 6\right]$

Integral values of x are $x = 1, 2$ and 6 .

23. $a \rightarrow p; b \rightarrow q; c \rightarrow q, s; d \rightarrow r, s$

Sol: a. $f(x) = \{(\operatorname{sgn} x)^{\operatorname{sgn} x}\}^n$

$$= \begin{cases} [(1)^1]^n, & x > 0 \\ [(-1)^{-1}]^n, & x < 0 \end{cases} = \begin{cases} 1, & x > 0 \\ -1, & x < 0 \end{cases}$$

Hence, $f(x)$ is an odd function.

$$\text{b. } f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$$

$$\text{or } f(-x) = \frac{-x}{e^{-x} - 1} - \frac{x}{2} + 1$$

$$= \frac{xe^x}{e^x - 1} - \frac{x}{2} + 1$$

$$= \frac{xe^x - x + x}{e^x - 1} - \frac{x}{2} + 1$$

$$= x + \frac{x}{e-1} - \frac{x}{2} + 1$$

$$= \frac{x}{e^x - 1} + \frac{x}{2} + 1$$

$$= f(x)$$

$$c. \quad f(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ 1, & \text{if } x \text{ is irrational} \end{cases}$$

$$\text{or } f(-x) = \begin{cases} 0, & \text{if } -x \text{ is rational} \\ 1, & \text{if } -x \text{ is irrational} \end{cases}$$

$$= \begin{cases} 0, & \text{if } x \text{ is rational} \\ 1, & \text{if } x \text{ is irrational} \end{cases}$$

$$= f(x)$$

$$d. \quad f(x) = \max\{\tan x, \cot x\}$$

from the graph of function, it can be verified that $f(x)$ is neither odd nor even.

$$\text{Also, } f(x + \pi) = \max\{\tan(x + \pi), \cot(x + \pi)\} \\ = \max\{\tan x, \cot x\}$$

Hence, $f(x)$ is periodic with period π .

24. 4

$$\text{Sol: } \lim_{x \rightarrow \infty} \left(\frac{x^3 + x^2 + x + \sin x}{x^2 + 2 \cos x} - a \sin x - bx + c \right) = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} \left(\frac{x + 1 + \frac{1}{x} + \frac{\sin x}{x^2}}{1 + \frac{2 \cos x}{x^2}} - a \sin x - bx + c \right) = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} (x + 1 - a \sin x - bx + c) = 4$$

$$\Rightarrow \lim_{x \rightarrow \infty} ((1 - b)x - a \sin x + 1 + c) = 4$$

$$\Rightarrow 1 - b = 0, a = 0 \text{ and } 1 + c = 4$$

$$\Rightarrow b = 1, a = 0 \text{ and } c = 3$$

25. 3

$$\text{Sol: } a. \text{ If } xy - \log y = 1 \text{ then } xy_1 + y = y_1 / y$$

$$\text{So } y^2 + (xy - 1)y_1 = 0$$

$$b. \quad y\sqrt{1-x^2} = \sin^{-1} x$$

$$\Rightarrow -\frac{yx}{\sqrt{1-x^2}} + y_1\sqrt{1-x^2} = \frac{1}{\sqrt{1-x^2}}$$

$$\text{So } -xy + y_1(1-x^2) = 1$$

$$c. \quad y^2 = 2x - x^2 \Rightarrow yy_1 = 1 - x$$

$$\Rightarrow y_1^2 + yy_2 = -1$$

$$\text{So, } \frac{(1-x)^2}{y^2} + yy_2 = -1$$

$$\Rightarrow y^3 y_2 = -[1+x^2-2x+y^2] = -1$$

$$\text{d. } y = e^{\sqrt{x}} + e^{-\sqrt{x}}$$

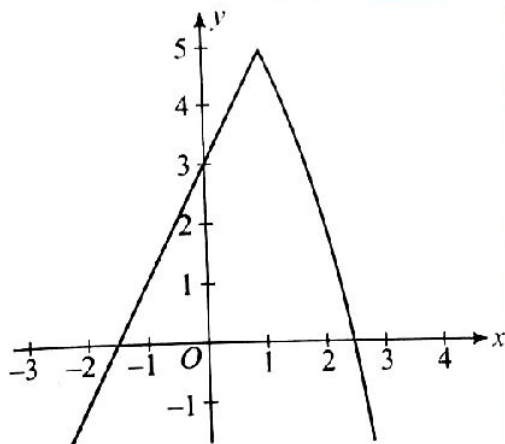
$$\Rightarrow y_1 = \frac{e^{\sqrt{x}} - e^{-\sqrt{x}}}{2\sqrt{x}}$$

$$\Rightarrow 4y_1^2 x = (e^{\sqrt{x}} - e^{-\sqrt{x}})^2 = y^{2-4}$$

26. 3

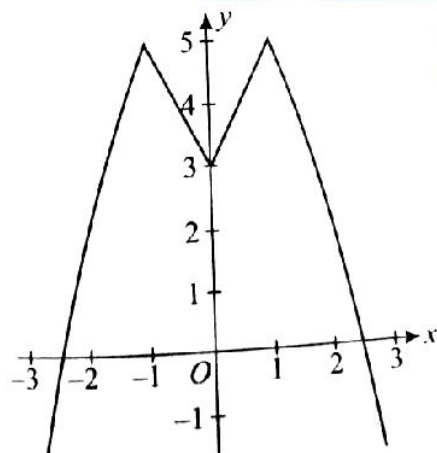
$$\text{Sol: } f(x) = \begin{cases} 2x+3, & x \leq 1 \\ -x^2+6, & x > 1 \end{cases}$$

Graph of $y = f(x)$ is as shown in the following figure.



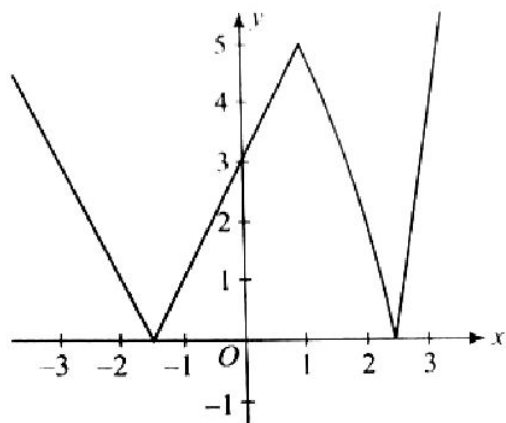
From the graph $f(x)$ has one point of non-differentiability.

Graph of $f(|x|)$ is as shown in the following figure.



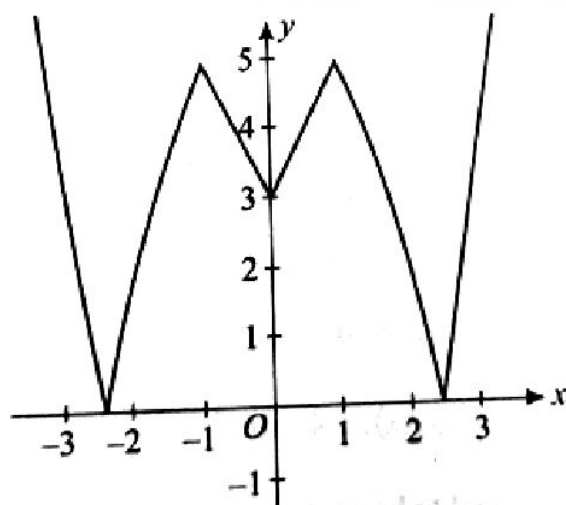
From the graph $f(|x|)$ has three points of non-differentiability.

Graph of $|f(x)|$ is as shown in the following figure.



From the graph $|f(x)|$ has three points of non – differentiability.

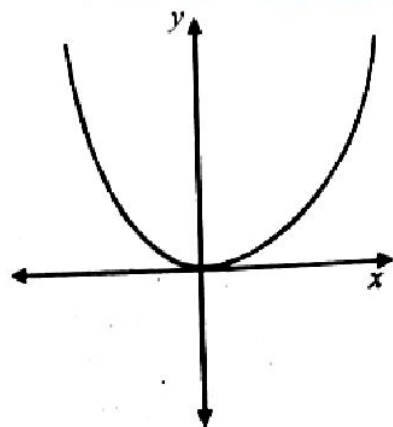
Graph of $|f(x)|$ is as shown in the following figure.



From the graph $|f(x)|$ has five points of non – differentiability.

27. D

Sol: $P(1): f_4(x) = \begin{cases} x^2, & x < 0 \\ e^{2x} - 1, & x \geq 0 \end{cases}$

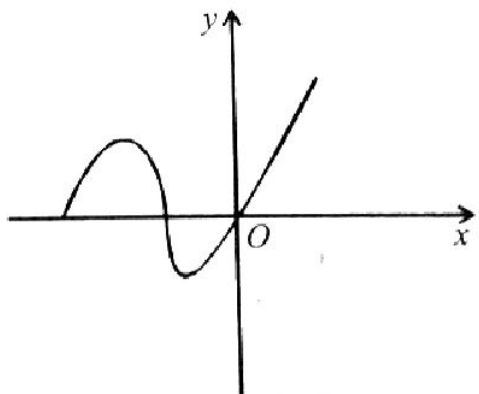


Range of $f_4 = [0, \infty)$

$\therefore f_4$ is onto

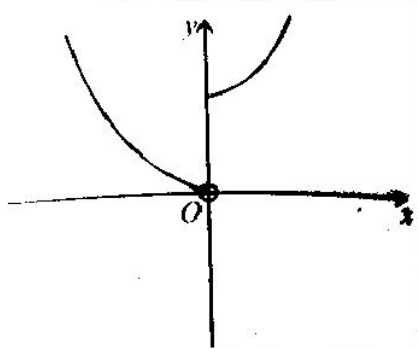
From graph f_4 is not one one.

$$Q(3): f_3(x) = \begin{cases} \sin x, & x < 0 \\ x, & x \geq 0 \end{cases}$$



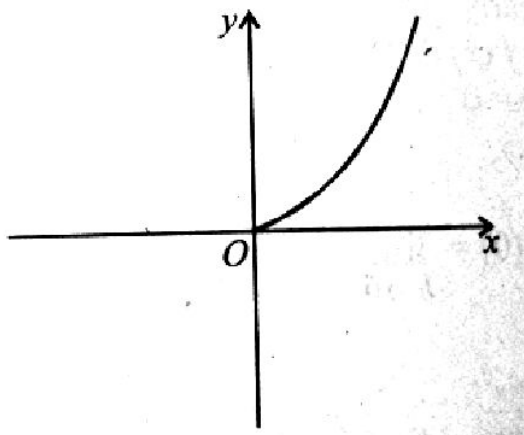
From graph f is differentiable but not one one.

$$R(2): f_2 \circ f_1(x) = \begin{cases} x^2, & x < 0 \\ e^{2x}, & x \geq 0 \end{cases}$$



From graph $f_2 \circ f_1$ is neither continuous nor one one.

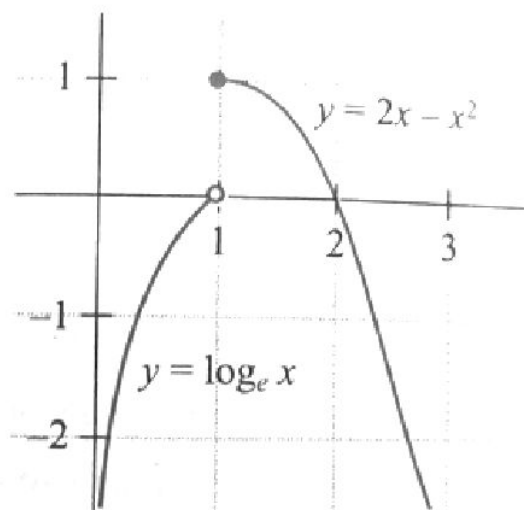
$$S(4): f_2(x) = x^2, x \in [0, \infty)$$



It is continuous and one one.

28. $a \rightarrow r; b \rightarrow s; c \rightarrow p; d \rightarrow q$

Sol:



b.s.
$$f(x) = \begin{cases} x-1, & x < 2 \\ 0, & x = 2 \\ \sin x, & x > 2 \end{cases}$$

$f(2) = 0, f(2^+) = \sin(2^+) > 0$, and $f(2^-) > 0$

Hence, $x = 2$ is point minima.

c.p.
$$f(x) = \begin{cases} 2x+3, & x < 0 \\ 5, & x = 0 \\ x^2+7, & x > 0 \end{cases}$$

$f(0^-) = 3, f(0) = 5, f(0^+) = 7$

Hence, $f(0^-) < f(0) < f(0^+)$.

Thus, $f(x)$ is increasing at $x = 0$.

d.p.
$$f(x) = \begin{cases} e^{-x}, & x < 0 \\ 0, & x = 0 \\ -\cos x, & x > 0 \end{cases}$$

$f(0) = 0, f(0^+) = -1, f(0^-) = 1$

Thus, $f(0^-) > f(0) > f(0^+)$

Hence, $f(x)$ is decreases at $x = 0$.

29. (B, C)

Sol: Case I : $x \neq m\pi$ (m is an integer)

$$\lim_{x \rightarrow \infty} \frac{1}{1+n \sin^2 nx} = \frac{1}{\infty} = 0$$

Case II: $x = m\pi$ (m is an integer)

$$\lim_{n \rightarrow \infty} \frac{1}{1+n \sin^2 nx} = \frac{1}{1} = 1$$

30. A,B,C,D

Sol: We have $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\tan^2 \{x\}}{(x^2 - [x]^2)} = \lim_{x \rightarrow 0^+} \frac{\tan^2 x}{x^2} = 1$ (1)

$$(\because x \rightarrow 0^+, [x] = 0 \Rightarrow \{x\} = x)$$

Also, $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \sqrt{\{x\} \cot \{x\}} = \sqrt{\cot 1}$ (2)

$$(\because x \rightarrow 0^-, [x] = -1 \Rightarrow \{x\} = x+1 \Rightarrow \{x\} \Rightarrow 1)$$

$$\text{Also, } \cot^{-1} \left(\lim_{x \rightarrow 0^-} f(x) \right)^2 = \cot^{-1}(\cot 1) = 1$$

$$\text{Also, } \tan^{-1} \left(\lim_{x \rightarrow 0^+} f(x) \right) = \tan^{-1} 1 = \frac{\pi}{4}$$

31. C

32. CD

Sol: $f(1^+) = \lim_{h \rightarrow 0} \frac{1 - (1+h)(1+h)}{h} \cos \frac{1}{h}$
 $= \lim_{h \rightarrow 0} \frac{-h^2 - 2h}{h} \cos \frac{1}{h} = \lim_{h \rightarrow 0} (-h - 2) \cos \frac{1}{h}$

Thus, $\lim_{x \rightarrow 1^+} f(x)$ does not exist.

$$f(1^-) = \lim_{h \rightarrow 0} \frac{1 - (1-h)(1+h)}{h} \cos \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{1 - (1-h^2)}{h} \cos \frac{1}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2}{h} \cos \frac{1}{h} = \lim_{h \rightarrow 0} h \cos \frac{1}{h} = 0$$

33. BC

Sol: $y = \cos^{-1} \left(\frac{2x}{1+x^2} \right)$

$$\text{Or } \frac{dy}{dx} = \frac{-1}{\sqrt{1 - \frac{4x^2}{(1+x^2)^2}}} \frac{d}{dx} \left(\frac{2x}{1+x^2} \right)$$

$$= - \frac{1+x^2}{\sqrt{(1-x^2)^2}} \frac{2(1+x^2) - 4x^2}{(1+x^2)^2}$$

$$= -2 \frac{(1+x^2)}{|1-x^2|} \frac{1-x^2}{(1+x^2)^2}$$

$$= -2 \left(\frac{1-x^2}{|1-x^2|} \right) \left(\frac{1}{1+x^2} \right)$$

$$= \begin{cases} \frac{2}{1+x^2}, & \text{if } |x| > 1 \\ \frac{-2}{1+x^2}, & \text{if } |x| < 1 \end{cases}$$

34. AC

Sol: $\frac{d}{dx}\{f_n(x)\} = \frac{d}{dx}\{e^{f_{n-1}(x)}\}$

$$= e^{f_{n-1}(x)} \frac{d}{dx}\{f_{n-1}(x)\} = f_n(x) \frac{d}{dx}\{f_n(x) \cdot e^{f_{n-2}(x)}\} \frac{d}{dx}\{f_{n-2}(x)\}$$

$$= f_n(x) \cdot \frac{d}{dx}\{e^{f_{n-2}(x)}\} = f_n(x) \cdot e^{f_{n-2}(x)} \frac{d}{dx}\{f_{n-2}(x)\}$$

$$= f_n(x) f_{n-1}(x) \frac{d}{dx}\{f_{n-2}(x)\}$$

$$= f_n(x) f_{n-1}(x) \dots f_2(x) \frac{d}{dx}\{f_1(x)\}$$

$$= f_n(x) f_{n-1}(x) \dots f_2(x) e^{f_0(x)} \frac{d}{dx}\{f_0(x)\}$$

Use $e^{f_0(x)} = f_1(x)$ and $f_0(x) = x$

35. BD

Sol: We have $g = \frac{1}{f}$

$$\therefore g' = \frac{-1}{f^2} f'$$

$$\text{Or } g'' = - \left[-\frac{2}{f^3} f'^2 + \frac{1}{f^2} f'' \right]$$

$$= \frac{2}{f^3} f'^2 - \frac{f''}{f^2}$$

$$\text{Or } \frac{f''}{f'} - \frac{g''}{g'} = \frac{f''}{f'} - \frac{\frac{2}{f^3} f'^2 - \frac{f''}{f^2}}{-\frac{1}{f^2} f'}$$

$$= \frac{f''}{f'} - \left(\frac{-2f'}{f} + \frac{f''}{f'} \right) = \frac{2f'}{f}$$

Also, $g \cdot f = 1$

$$\text{Or } g'f + gf' = 0$$

$$\therefore \frac{f''}{f'} - \frac{g''}{g'} = -\frac{2g'}{g}$$

36. AD

Sol: Let f and g be maximum at c_1 and c_2 respectively,

$$c_1, c_2 \in [0, 1]$$

$$\text{Then, } f(c_1) = g(c_2)$$

$$\text{Let } h(x) = f(x) - g(x)$$

$$\text{Then, } h(c_1) = f(c_1) - g(c_1) > 0$$

$$\text{And } h(c_2) = f(c_2) - g(c_2) < 0$$

$$\therefore h(x) = 0 \text{ has atleast one root in } [c_1, c_2]$$

$$\text{i.e. } f(c) = g(c) \text{ for } c \in [c_1, c_2]$$

Which shows that options (a) and (d) are correct.

37. BD

$$\text{Sol: Given : } f(x) = \begin{cases} a_n + \sin \pi x, & x \in [2n, 2n+1] \\ b_n + \cos \pi x, & x \in (2n-1, 2n) \end{cases}$$

$\therefore f$ is continuous for all n

$$\therefore \text{At } x = 2n, \text{LHL} = \text{RHL} = f(2n)$$

$$\Rightarrow b_n + \cos 2\pi n = a_n + \sin 2\pi n = a_n + \sin 2\pi n$$

$$\Rightarrow b_n + 1 = a_n \Rightarrow a_n - b_n = 1, \therefore \text{option (b) is correct.}$$

$$\text{Also at } x = 2n+1, \text{LHL} = \text{RHL} = f(2n+1)$$

$$\Rightarrow \lim_{h \rightarrow 0} a_n + \sin \pi(2n+1-h)$$

$$= \lim_{h \rightarrow 0} b_{n+1} + \cos \pi(2n+1-h) = a_n + \sin(2n+1)\pi$$

$$\Rightarrow a_n = b_{n+1} - 1 = a_n \Rightarrow a_n - b_{n+1} = -1$$

$\therefore$ option (c) is incorrect.

$$\Rightarrow a_{n-1} - b_n = -1, \therefore \text{option (d) is correct.}$$

38. AD

$$\text{Sol: } f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} -g(x), & x < 0 \\ 0, & x = 0 \\ g(x), & x > 0 \end{cases}$$

$$f'(x) = \begin{cases} -g'(x), & x < 0 \\ 0, & x = 0 \\ g'(x), & x > 0 \end{cases}$$

$$\therefore \text{Lf}'(0) = -g'(0) = 0 \text{ and } \text{Rf}'(0) = g'(0) = 0$$

$\therefore f$ is differentiable at $x = 0$

$$h(x) = e^{|x|} = \begin{cases} e^{-x}, & x < 0 \\ e^x, & x \geq 0 \end{cases}$$

$$h'(x) = \begin{cases} -e^{-x}, & x < 0 \\ e^x, & x \geq 0 \end{cases}$$

$$\therefore \text{Lh}'(0) = -1, \text{Rh}'(0) = 1$$

$\therefore h$ is not differentiable at $x = 0$

$$f \circ h(x) = f(h(x)) = f(e^{|x|})$$

$$= \begin{cases} g(e^{-x}) & \text{if } x < 0 \\ g(1) & \text{if } x = 0 \\ g(e^x) & \text{if } x > 0 \end{cases}$$

$$f'[h(x)] = \begin{cases} -g'(e^{-x}) \cdot e^{-x}, & x < 0 \\ 0, & x = 0 \\ g'(e^x) e^x, & x > 0 \end{cases}$$

$$Lf'(h(0)) = -g'(1), Rf'(h(0)) = g'(1)$$

$$\therefore g'(1) \neq 0, \therefore Lf'(h(0)) \neq Rf'(h(0))$$

$\therefore f \circ h$ is not differentiable at $x = 0$.

$$h \circ f(x) = \begin{cases} e^{|f(x)|}, & x \neq 0 \\ 1, & x = 0 \end{cases}$$

$$Lh'(f(0)) = \lim_{k \rightarrow 0} \frac{h(f(0)) - h(f(0-k))}{k}$$

$$= \lim_{k \rightarrow 0} \frac{1 - e^{|g(-k)|}}{k} = \lim_{k \rightarrow 0} \frac{1 - e^{|g(-k)|}}{|g(-k)|} \times \frac{|g(-k)|}{k}$$

$$= 1 \times 0 = 0 \left(\because g'(0) = 0 \Rightarrow \lim_{k \rightarrow 0} \frac{g(-k)}{k} = \lim_{k \rightarrow 0} \frac{g(k)}{k} = 0 \right)$$

$$Rh'(f(0)) = \lim_{k \rightarrow 0} \frac{h(f(0+k)) - h(f(0))}{k}$$

$$= \lim_{k \rightarrow 0} \frac{e^{|g(k)|} - 1}{k} = \lim_{k \rightarrow 0} \frac{e^{|g(k)|} - 1}{|g(k)|} \times \frac{|g(k)|}{k} = 0$$

$$\therefore Lh'(f(0)) = Rh'(f(0)) = 0$$

$\therefore h \circ f$ is differentiable at $x = 0$

39. BC

Sol: Given: $f(x) = x^3 + 3x + 2 \Rightarrow f'(x) = 3x^2 + 3$

$$\therefore f(0) = 2, f(1) = 6, f(2) = 16, f(3) = 38, f(6) = 236$$

$$\text{Also given } g(f(x)) = x \Rightarrow g(2) = 0, g(6) = 1, g(16) = 2,$$

$$g(38) = 3, g(236) = 6$$

$$(a) g(f(x)) = x \Rightarrow g'(f(x)) \cdot f'(x) = 1$$

$$\text{For } g'(2), f(x) = 2 \Rightarrow x = 0$$

$$\text{On putting } x = 0, \text{ we get } g'(f(0))f'(0) = 1$$

$$\Rightarrow g'(2) = \frac{1}{3}$$

$$(b) h(g(g(x))) = x \Rightarrow h'(g(g(x))) \cdot g'(g(x)) \cdot g'(x) = 1$$

$$\text{For } h'(1), \text{ we need } g(g(x)) = 1$$

$$\Rightarrow g(x) = 6 \Rightarrow x = 236$$

On putting $x = 236$, we get

$$h'[g(g(236))] = \frac{1}{g'(g(236))g'(236)}$$

$$\Rightarrow h'(g(6)) = \frac{1}{g'(6)g'(236)}$$

$$\begin{aligned}\Rightarrow h'(1) &= \frac{1}{g'(f(1))g'(f(6))} \\ &= f'(1)f'(6) = 6 \times 111 = 666\end{aligned}$$

(c) $h[g(g(x))] = x$

For $h(0)$, $g(g(x)) = 0 \Rightarrow g(x) = 2 \Rightarrow x = 16$

On Putting $x = 16$, we get

$$h(g(g(16))) = 16 \Rightarrow h(0) = 16$$

(d) $h[g(g(x))] = x$

For $h(g(3))$, we need $g(x) = 3 \Rightarrow x = 38$

On putting $x = 38$, we get

$$h[g(g(38))] = 38 \Rightarrow h(g(3)) = 38$$

40. AC

Sol: Let length and breadth of rectangular sheet be $8x$, and $15x$ respectively. Also let y be the length of square cut off from each corner.

$$\therefore \text{Volume of box} = (8x - 2y)(15x - 2y)y$$

$$\Rightarrow V = 120x^2y - 46xy^2 + 4y^3$$

$$\frac{dV}{dy} = 120x^2 - 92xy + 12y^2$$

$$\text{Now, } 4y^2 = 100 \Rightarrow y = 5$$

$$\text{Put } \frac{dV}{dy} = 0 \text{ for maximum value of } V.$$

$$\Rightarrow 30x^2 - 115x + 75 = 0 \Rightarrow x = 3, \frac{5}{6}$$

For $x = 3$, sides are 45 and 24.

41. BC

Sol: Let $h(x) = f(x) - 3g(x)$

$$h(-1) = h(0) = h(2) = 3$$

$\therefore$ By Rolle's theorem $h'(x) = 0$ has atleast one solution in $(-1, 0)$ and atleast one solution in $(0, 2)$

But $h''(x)$ never vanishes in $(-1, 0)$ and $(0, 2)$ therefore $h'(x) = 0$ should have exactly one solution in each of the two intervals $(-1, 0)$ and $(0, 2)$.

42. A, B, C, D

Sol: a. $y^2 = 4ax$ or $m_1 = y' = \frac{2a}{y}$

$$y = e^{-x}/2a \text{ or } m_2 = y' = -\frac{1}{2a}e^{-x/2a} = -\frac{1}{2a}y$$

$$m_1 m_2 = -1.$$

Hence, orthogonal.

b. $y^2 = 4ax$

or $y' = \frac{4a}{2y_1} = \frac{2a}{y_1}$, [Not defined at (0, 0)]

$$x^2 = 4ay$$

Or $y' = \frac{2x_1}{4a} = \frac{x_1}{2a} = 0$ at (0, 0)

Therefore, the two curves are orthogonal at (0, 0)

c. $xy = a^2, x^2 - y^2 = b^2$

$$\therefore m_1 = y' = -\frac{y}{x}$$

And $m_2 = y' = \frac{x}{y}$

$$m_1 m_2 = -1$$

Hence, orthogonal.

d. $y = ax \Rightarrow y' = a = m_1$

$$x^2 + y^2 = c^2 \Rightarrow y' = -\frac{x_1}{y_1} = m_2$$

$$m_1 m_2 = -\frac{ax_1}{y_1} = -\frac{y_1}{y_1} = -1$$

Hence, orthogonal.

42. AB

Sol: Let P (x, y) be a point on the curve $\ln(x^2 + y^2) = c \tan^{-1} \frac{y}{x}$.

Differentiating both sides with respect to x, we get

$$\frac{2x + 2yy'}{(x^2 + y^2)} = \frac{c(xy' - y)}{(x^2 + y^2)} \text{ or } y' = \frac{2x + cy}{cx - 2y} = m_1 \text{ (say)}$$

Slope of OP = $\frac{y}{x} = m_2$ (say (where O is origin))

Let the angle between the tangents at P and OP be θ . Then,

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \left| \frac{\frac{2x + cy}{cx - 2y} - \frac{y}{x}}{1 + \frac{2xy + cy^2}{cx^2 - 2xy}} \right| = \frac{2}{c}$$

$$\text{Or } \theta = \tan^{-1} \left(\frac{2}{c} \right)$$

Which is independent of x and y.

43. ABC

Sol: (A) Let $f(x) = e^x \cos x - 1$

$$\therefore f'(x) = e^x (\cos x - \sin x) = 0$$

$$\therefore \tan x = 1, \text{ which has a root between two roots of } f(x) = 0$$

(B) Let $f(x) = e^x \sin x - 1$

$$\therefore f'(x) = e^x (\sin x + \cos x) = 0$$

$$\therefore \tan x = -1, \text{ which has a root between two roots of } f(x) = 0$$

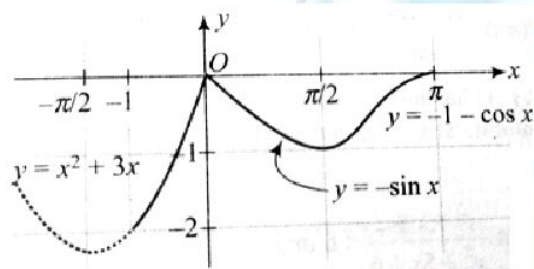
(C) Let $f(x) = e^{-x} - \cos x$

$$\therefore f'(x) = -e^{-x} + \sin x = 0$$

$$\therefore e^{-x} = \sin x, \text{ which has a root between two roots of } f(x) = 0$$

44. A, B, C, D

Sol:



From the graph, global minimum value is $f(-1) = -2$ and global maximum value is $f(0) = f(\pi) = 0$.

45. A, D

Sol: We have

$$f(x) = 2x + \cot^{-1} x + \log(\sqrt{1+x^2} - x)$$

$$\therefore f'(x) = 2 - \frac{1}{1+x^2} + \frac{1}{\sqrt{1+x^2} - x} \left(\frac{x}{\sqrt{1+x^2}} - 1 \right)$$

$$= \frac{1+2x^2}{1+x^2} - \frac{1}{\sqrt{1+x^2}} = \frac{1+2x^2}{1+x^2} - \frac{\sqrt{1+x^2}}{1+x^2}$$

$$= \frac{x^2 + \sqrt{1+x^2}(\sqrt{1+x^2} - 1)}{1+x^2} > 0 \text{ for all } x$$

Hence, $f(x)$ is an increasing function in $(-\infty, \infty)$ and, in particular, in $(0, \infty)$.

46. BC

Sol: Let $G(x) = e^x \cos x f(x)$

$$\therefore G'(x) = e^x \cos x f'(x) + e^x (\cos x - \sin x) f(x)$$

$$= e^x (\cos x f'(x) + (\cos x - \sin x) f(x)) \leq 0 \text{ (as given that } \cos x f'(x) \leq (\sin x - \cos x) f(x))$$

$$\therefore G(x) \text{ is a non increasing function.}$$

$$\therefore G(6) = e^6 \cos 6 f(6) \leq 0$$

$$\therefore f(6) \leq 0$$

But given that $f(x)$ is non-negative

$$\therefore f(6) = 0$$

With similar reasons $f(5), f(7) = 0$

$$\text{Thus } x^2 - 3x + 2 + f(7) = 0$$

Or $x^2 - 3x + 2 = 0$ has 2 distinct solutions

$$\lim_{x \rightarrow 6} \frac{f(x) - \sin(\pi x)}{(x - 6)}$$

$$= \lim_{x \rightarrow 6} \frac{f'(x) - \pi \cos(\pi x)}{1} \quad (\text{applying L' Hospital Rule})$$

$$= f'(6) - \pi < 0 \quad (\text{as } f'(6) \leq 0)$$

47. AD

$$\text{Sol: } f(x) = \frac{x^2}{2 - 2 \cos x} = \frac{x^2}{4} \operatorname{cosec}^2 \frac{x}{2}$$

$$\Rightarrow f'(x) = \frac{x}{2} \operatorname{cosec}^2 \frac{x}{2} \cot \frac{x}{2} \left(\tan \frac{x}{2} - \frac{x}{2} \right)$$

$$\text{For } 0 < x < 1, \tan \frac{x}{2} > \frac{x}{2}$$

$$\Rightarrow f'(x) > 0$$

So, $f(x)$ is increasing.

$$g(x) = \frac{x^2}{6x - 6 \sin x}$$

$$\Rightarrow g'(x) = \frac{1}{6} \frac{2x(x - \sin x) - x^2(1 - \cos x)}{(x - \sin x)^2}$$

$$= \frac{x}{6} \frac{x - 2 \sin x + x \cos x}{(x - \sin x)^2}$$

$$\text{Let } h(x) = x - 2 \sin x + x \cos x$$

$$h'(x) = 1 - \cos x + \cos x - x \sin x$$

$$= 1 - \cos x - x \sin x$$

$$= 2 \sin^2 \frac{x}{2} - 2x \sin \frac{x}{2} \cos \frac{x}{2}$$

$$= 2 \sin \frac{x}{2} \cos \frac{x}{2} \left(\tan \frac{x}{2} - x \right)$$

$$\text{For } 0 < x < 1, \tan \frac{x}{2} < x$$

$$\therefore h'(x) < 0$$

So, $h(x)$ is decreasing.

$$\text{For } x > 0, h(x) < h(0)$$

$$\therefore h(x) < 0$$

So, $h(x)$ is decreasing

For $x > 0, h(x) < h(0)$

$$\therefore h(x) < 0 \quad \therefore g'(x) < 0$$

So, $g(x)$ is decreasing.

48. ABC

Sol: $f(-x) = -x \cos(x) - \sin(-x) = -x \cos x + \sin x = -f(x)$

$\therefore f(x)$ is odd function.

$$f'(x) = -x \sin x + \cos x - \cos x = -x \sin x$$

$$f'(x) = 0 \Rightarrow x = n\pi, n \in \mathbb{Z}.$$

$$f''(x) = -x \cos x - \sin x$$

$$f''(0) = 0$$

$\therefore x = 0$ is point of inflection.

$$f'(0) = (-)(-)(-) < 0$$

$$f'(0^+) = (-)(+)(+) < 0$$

Thus, at $x = 0$ $f(x)$ decreases.

$$f''(\pi) = \pi > 0$$

$\therefore x = \pi$ is point of minima.

49. ABCD

Sol: Given function is discontinuous when $a + \sin \pi x = \pm 1$.

Now, if $a = 1$, then $\sin \pi x = 0$ or $x = 1, 2, 3, 4, 5$.

If $a = 3$, then $\sin \pi x = -2$, not possible.

If $a = 0.5$, then $\sin \pi x = 0.5$

Therefore, x has 6 value, 2 each for one cycle of period 2.

$$\text{If } a = 0, \text{ then } \sin \pi x = \pm 1 \text{ or } x = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}$$

Hence, all the options are correct.

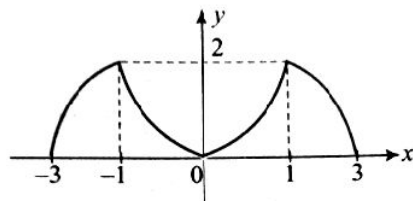
50. ABC

Sol: a. $-1 \leq |x| - 1 \leq 4$

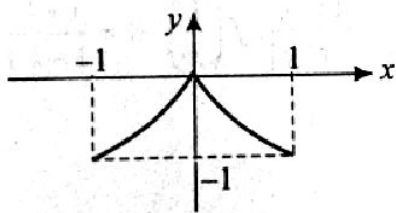
$$\Rightarrow 0 \leq |x| \leq 5 \Rightarrow -5 \leq x \leq 5$$

b. $f(|x|+1)$

range of $f(|x|+1)$ is $[0, 2]$



3. range of $f(-|x|)$ is $[-1, 0]$



$$4. -1 \leq |x| \leq 4$$

$$\Rightarrow x \in [-4, 4]$$

51. ACD

$$\text{Sol: } f^2(x) = f\left(\frac{3}{4}x+1\right) = \frac{3}{4}\left(\frac{3}{4}x+1\right) + 1 = \left(\frac{3}{4}\right)^2 x + \frac{3}{4} + 1 \quad (1)$$

$$f^3(x) = f\{f^2(x)\} = \frac{3}{4}\{f^2(x)+1\}$$

$$= \frac{3}{4}\left\{\left(\frac{3}{4}\right)^2 x + \frac{3}{4} + 1\right\} + 1$$

$$= \left(\frac{3}{4}\right)^3 x + \left(\frac{3}{4}\right)^2 + \frac{3}{4} + 1$$

$$\therefore f^n(x) = \left(\frac{3}{4}\right)^n x + \left(\frac{3}{4}\right)^{n-1} + \left(\frac{3}{4}\right)^{n-2} + \dots + \left(\frac{3}{4}\right) + 1$$

$$= \left(\frac{3}{4}\right)^n x + \frac{1 - \left(\frac{3}{4}\right)^n}{1 - \frac{3}{4}}$$

$$\therefore \lambda = \lim_{n \rightarrow \infty} f^n(x) = 0 + 4 = 4$$

52. A, B, C

$$\text{Sol: } f(\alpha) = \lim_{x \rightarrow 2} (\sin^x \alpha + \cos^x \alpha)^{\frac{1}{x-2}} \quad (1^\infty \text{ form})$$

$$= \begin{cases} e^{\lim_{x \rightarrow 2} \frac{\sin^x \alpha + \cos^x \alpha - 1}{x-2}}, & \alpha \in \left(0, \frac{\pi}{2}\right) \\ 1, & \alpha = 0, \frac{\pi}{2} \end{cases}$$

$$\begin{aligned} \text{Now } e^{\lim_{x \rightarrow 2} \frac{\sin^x \alpha + \cos^x \alpha - 1}{x-2}} &= e^{\lim_{x \rightarrow 2} \frac{\sin^x \alpha + \cos^x \alpha - \sin^2 \alpha - \cos^2 \alpha}{x-2}} \\ &= e^{\lim_{x \rightarrow 2} \frac{\sin^2 \alpha (\sin^{x-2} \alpha - 1) + \cos^2 \alpha (\cos^{x-2} \alpha - 1)}{x-2}} \\ &= e^{\sin^2 \alpha \log_e \sin \alpha + \cos^2 \alpha \log_e \cos \alpha} \\ &= e^{\log_e (\sin \alpha) \sin^2 \alpha + \log_e (\cos \alpha) \cos^2 \alpha} \\ &= (\sin \alpha)^{\sin^2 \alpha} (\cos \alpha)^{\cos^2 \alpha} \end{aligned}$$

$$\therefore f(x) = \begin{cases} (\cos \alpha) \cos^2 \alpha (\sin \alpha)^{\sin^2 \alpha}, & \alpha \in \left(0, \frac{\pi}{2}\right) \\ 1, & \alpha = 0, \frac{\pi}{2} \end{cases}$$

53. B

Sol: $\lim_{x \rightarrow 1} \frac{(1-x)(1-x^2)\dots(1-x^{2n})}{\{(1-x)(1-x^2)\dots(1-x^n)\}^2}$

$$\lim_{x \rightarrow 1} \frac{\left(\frac{1-x}{1-x}\right)\left(\frac{1-x^2}{1-x}\right)\dots\left(\frac{1-x^{2n}}{1-x}\right)}{\left(\left(\frac{1-x}{1-x}\right)\left(\frac{1-x^2}{1-x}\right)\dots\left(\frac{1-x^n}{1-x}\right)\right)^2}$$

$$= \frac{1 \times 2 \times 3 \dots (2n)}{(1 \times 2 \times 3 \dots n)^2} = \frac{(2n)!}{n!n!} = {}^{2n}C_n$$

54. D

Sol: $\lim_{x \rightarrow 0} \frac{1 - \cos x \sqrt{\cos 2x}}{x^2}$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos x \sqrt{\cos 2x})(1 + \cos x \sqrt{\cos 2x})}{x^2(1 + \cos x \sqrt{\cos 2x})}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x (\cos 2x)}{2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{1 - \cos^2 x (2 \cos^2 x - 1)}{2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{(1 - \cos^2 x)(1 + 2 \cos^2 x)}{2x^2}$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x (1 + 2 \cos^2 x)}{2x^2}$$

$$= \frac{3}{2}$$

55. B

Sol: $\cos(\tan x) - \cos x = 2 \sin\left(\frac{x + \tan x}{2}\right) \sin\left(\frac{x - \tan x}{2}\right)$

Or $\lim_{x \rightarrow 0} \frac{\cos(\tan x) - \cos x}{x^4}$

$$= \lim_{x \rightarrow 0} \frac{2 \sin\left(\frac{x + \tan x}{2}\right) \sin\left(\frac{x - \tan x}{2}\right)}{x^4}$$

$$\begin{aligned}
 &= \lim_{x \rightarrow 0} \frac{2 \sin\left(\frac{x + \tan x}{2}\right) \sin\left(\frac{x - \tan x}{2}\right)}{x^4 \left(\frac{x + \tan x}{2}\right) \left(\frac{x - \tan x}{2}\right)} \left(\frac{x^2 - \tan^2 x}{4}\right) \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{x^2 - \tan^2 x}{x^4} \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{x^2 - \left(x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots\right)^2}{x^4} \\
 &= \frac{1}{2} \lim_{x \rightarrow 0} \frac{1}{x^2} \left(1 - \left(1 + \frac{x^2}{3} + \frac{2}{15}x^4 + \dots\right)^2\right) = -\frac{1}{3}
 \end{aligned}$$

56. B

57. A

Sol: $S = \sum_{n=1}^9 \frac{2}{4(n^2 + 3n + 2)} = \frac{1}{2} \sum_{n=1}^9 \left(\frac{1}{n+1} - \frac{1}{n+2}\right)$

$$S = \frac{1}{2} \left(\frac{1}{2} - \frac{1}{11}\right) = \frac{9}{44}$$

58. C

Sol: $\frac{d}{dx} \left(\frac{g(x)}{g(g(x))} \right) = \frac{g(g(x)) \cdot g'(x) - g(x) \cdot g'(g(x)) \cdot g'(x)}{(g(g(x)))^2}$

Now, $f(0) = 4$

$$\Rightarrow f^{-1}(4) = g(4) = 0 \quad \text{And } g'(4) = \frac{1}{f'(0)} = \frac{1}{3}$$

$$\text{Also, } f(-1) = 0 \quad \Rightarrow f^{-1}(0) = g(0) = -1$$

$$\begin{aligned}
 \therefore \frac{d}{dx} \left(\frac{g(x)}{g(g(x))} \right)_{x=4} &= \frac{g(g(4))g'(4) - g(4)g'(g(4))g'(4)}{(g(4))^2} \\
 &= \frac{g(0)g'(4) - 0}{(g(4))^2}
 \end{aligned}$$

$$= \frac{g(0) \cdot \frac{1}{f'(0)} - 0}{(g(0))^2}$$

$$= \frac{-1 \times \frac{1}{3}}{1}$$

$$= \frac{-1}{3}$$

59. B

Sol: $y = 2 \cos x \cos 3x = \cos 4x + \cos 2x$

$$\therefore \frac{d^{20}y}{dx^{20}} = 4^{20} \cos 4x + 2^{20} \cos 2x$$

60. A

Sol: we have

$$y^{1/m} = (x + \sqrt{1+x^2})$$

$$\text{Or } y = (x + \sqrt{1+x^2})^m$$

$$\text{Or } \frac{dy}{dx} = m(x + \sqrt{1+x^2})^{m-1} \left(1 + \frac{x}{\sqrt{x^2+1}}\right)$$

$$= m \frac{(x + \sqrt{1+x^2})^m}{\sqrt{1+x^2}}$$

$$= \frac{my}{\sqrt{1+x^2}}$$

$$\text{Or } y_1^2(1+x^2) = m^2 y^2$$

$$\text{Or } 2y_1 y_2(1+x^2) + 2xy_1^2 = 2m^2 yy_1$$

$$\text{Or } y_2(1+x^2) + xy_1 = m^2 y$$

61. A

Sol: We know that when $x \rightarrow 0^-$, then $\{x\} = 1-h$ where $h \rightarrow 0$

$$\text{LHL} = \lim_{h \rightarrow 0} \frac{\cos^{-1}(1-(1-h)^2) \sin^{-1}(h)}{(1-h)(1-(1-h)^2)}$$

$$\lim_{h \rightarrow 0} \frac{\cos^{-1}(1-(1-h)^2) \sin^{-1}(h)}{(1-h)h(2-h)} = \frac{\frac{\pi}{2} \times 1}{1 \times 2} = \frac{\pi}{4}$$

And when $x \rightarrow 0^+$ then $\{x\} = h$ where $h \rightarrow 0$

$$\text{RHL} = \lim_{h \rightarrow 0} \frac{\cos^{-1}(1-h^2) \sin^{-1}(1-h)}{h-h^3}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1} \sqrt{1-(1-h^2)^2} \sin^{-1}(1-h)}{h(1-h^2)}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(h\sqrt{2-h^2}) \sin^{-1}(1-h)}{h(1-h^2)}$$

$$= \lim_{h \rightarrow 0} \frac{\sin^{-1}(h\sqrt{2-h^2})}{h\sqrt{2-h^2}} \times \frac{h\sqrt{2-h^2}}{h(1-h^2)} \times \sin^{-1}(1-h)$$

$$= 1 \times \frac{\sqrt{2}}{1} \times \frac{\pi}{2} = \frac{\pi}{\sqrt{2}}$$

LHL $\neq$ RHLSo, $f(x)$ is discontinuous at $x = 0$.

62. D

Sol: $f(x) = \begin{cases} -1, & -2 \leq x < 0 \\ x^2 - 1, & 0 \leq x \leq 2 \end{cases}$

Then, $f(|x|) = \begin{cases} -1, & -2 \leq |x| < 0 \\ |x|^2 - 1, & 0 \leq |x| \leq 2 \end{cases}$

$\Rightarrow f(|x|) = x^2 - 1, -2 \leq x \leq 2$

$\Rightarrow g(x) = \begin{cases} 1 + x^2 - 1, & -2 \leq x < 0 \\ (x^2 - 1) + |x^2 - 1|, & 0 \leq x \leq 2 \end{cases}$

$= \begin{cases} x^2, & -2 \leq x < 0 \\ 0, & 0 \leq x < 1 \\ 2(x^2 - 1), & 1 \leq x \leq 2 \end{cases}$

$g'(0) = 0, g'(0^+) = 0, g'(1^-) = 0, g'(1^+) = 4$

$\Rightarrow g(x)$ is non-differentiable at $x = 1$

$\Rightarrow g(x)$ is not differentiable at one point

63. D

Sol: Let $u = \tan^{-1} \left(\frac{\sqrt{1+x^2}-1}{x} \right)$

Put $x = \tan \theta \Rightarrow \theta = \tan^{-1} x$

$\therefore u = \tan^{-1} \left(\frac{\sec \theta - 1}{\tan \theta} \right) = \tan^{-1} \left(\tan \frac{\theta}{2} \right)$

$= \frac{\theta}{2} = \frac{1}{2} \tan^{-1} x$

$\therefore \frac{du}{dx} = \frac{1}{2} \times \frac{1}{(1+x^2)}$

Let $v = \tan^{-1} \left(\frac{2x\sqrt{1-x^2}}{1-2x^2} \right)$

Put $x = \sin \phi \Rightarrow \phi = \sin^{-1} x$

$v = \tan^{-1} \left(\frac{2 \sin \phi \cos \phi}{\cos 2\phi} \right) = \tan^{-1}(\tan 2\phi) = 2\phi = 2 \sin^{-1} x$

$\frac{dv}{dx} = 2 \frac{1}{\sqrt{1-x^2}}$

$\frac{du}{dv} = \frac{du/dx}{dv/dx} = \frac{\sqrt{1-x^2}}{4(1+x^2)}$

$\therefore \left(\frac{du}{dv} \right)_{\left(x=\frac{1}{2} \right)} = \frac{\sqrt{3}}{10}$

64. D

Sol: $y = \{x + \sqrt{x^2 - 1}\}^{15} + \{x - \sqrt{x^2 - 1}\}^{15}$

Differentiate w.r.t. 'x'

$$\frac{dy}{dx} = 15(x + \sqrt{x^2 - 1})^{14} \left[1 + \frac{x}{\sqrt{x^2 - 1}} \right] + 15(x - \sqrt{x^2 - 1})^{14} \left(1 - \frac{x}{\sqrt{x^2 - 1}} \right)$$

$$\Rightarrow \frac{dy}{dx} = \frac{15}{\sqrt{x^2 - 1}} \cdot y$$

$$\Rightarrow \sqrt{x^2 - 1} \cdot \frac{dy}{dx} = 15y$$

Again differentiating both sides w.r.t. x

$$\frac{x}{\sqrt{x^2 - 1}} \cdot \frac{dy}{dx} + \sqrt{x^2 - 1} \frac{d^2y}{dx^2} = 15 \frac{dy}{dx}$$

$$\Rightarrow x \frac{dy}{dx} + (x^2 - 1) \frac{d^2y}{dx^2}$$

$$= 15\sqrt{x^2 - 1} \cdot \frac{15}{\sqrt{x^2 - 1}} \cdot y = 225y$$

65. B

Sol: We have $x = \phi(t)$, $y = \psi(t)$. Therefore,

$$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{\psi'}{\phi'}$$

$$\text{Or } \frac{d^2y}{dx^2} = \frac{d}{dx} \left(\frac{\psi'}{\phi'} \right) = \frac{d}{dt} \left(\frac{\psi'}{\phi'} \right) \frac{dt}{dx}$$

$$= \frac{\phi' \psi'' - \psi' \phi''}{\phi'^2} \cdot \frac{1}{\phi'} = \frac{\phi' \psi'' - \psi' \phi''}{\phi'^3}$$

66. 3

Sol: Given that $\cos \frac{x}{2} \cos \frac{x}{2^2} \cos \frac{x}{2^3} \dots \cos \frac{x}{2^n} = \frac{\sin x}{2^n \sin \left(\frac{x}{2^n} \right)}$ (1)

Taking logarithm to the base e on both sides of equation (1) and then differentiating w.r.t. x, we get

$$\sum_{n=1}^n \frac{1}{2^n} \tan \frac{x}{2^n} = \left(\frac{1}{2^n} \cot \frac{x}{2^n} - \cot x \right)$$

$$\therefore \lim_{n \rightarrow \infty} \sum_{n=1}^n \frac{1}{2^n} \tan \frac{x}{2^n} = \lim_{n \rightarrow \infty} \left(\frac{1}{x} \times \frac{\frac{x}{2^n}}{\tan \frac{x}{2^n}} - \cot x \right) = \left(\frac{1}{x} - \cot x \right)$$

Thus, we have $f(x) = \begin{cases} \frac{1}{x} - \cot x, & x \in (0, \pi) - \left\{\frac{\pi}{2}\right\} \\ \frac{2}{\pi}, & x = \frac{\pi}{2} \end{cases}$

Clearly, $\lim_{x \rightarrow \frac{\pi}{2}} f(x) = \lim_{x \rightarrow \frac{\pi}{2}} \left(\frac{1}{x} - \cot x \right) = \frac{2}{\pi} = f\left(\frac{\pi}{2}\right)$

Hence, $f(x)$ is continuous at $x = \frac{\pi}{2}$

67. C

Sol: $2 + 10 \sin x \in [2, 12]$

Clearly, $f(x)$ is discontinuous where $2 + 10 \sin x = 3, 4, 5, \dots, 12$, i.e., at 10 points.

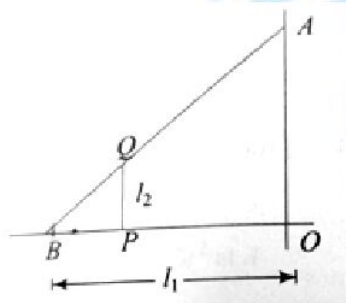
Number of points of non-differentiability of $g(x) = (x-1)(x-2) \dots (x-2m)$ is $(2m-2)$ as $g(x)$ is differentiable at $x=1$ and $x=2$.

$$\therefore 2m - 2 = 10$$

$$\therefore m = 6$$

68. B

Sol:



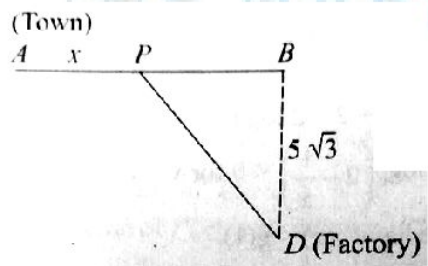
Let $BP = x$. From similar triangle property, we get

$$\frac{AO}{l_1} = \frac{l_2}{x} \text{ or } AO = \frac{l_1 l_2}{x} \text{ or } \frac{d(AO)}{dt} = -\frac{l_1 l_2}{x^2} \frac{dx}{dt}$$

$$\text{When } x = \frac{l_1}{2}, \frac{d(AO)}{dt} = -\frac{2l_2}{5} \text{ m/s.}$$

69. A

Sol:



Let the charges for railway line be kRs / km .

$$\text{Now, total freight charges, } T = kx + 2k\sqrt{(20-x)^2 + 75}$$

$$\text{Let } \frac{dT}{dx} = 0$$

$$\text{Or } k + 2k \frac{2(x-20)}{2\sqrt{(x-20)^2 + 75}} = 0$$

$$\text{Or } 4(x-20)^2 = 75 + (x-20)^2$$

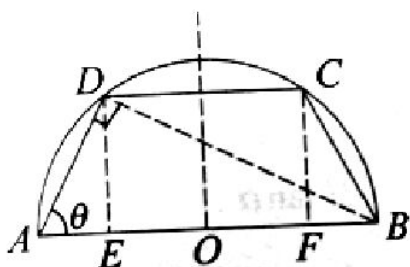
$$\text{Or } (x-20)^2 = 25$$

$$\text{Or } x = 25, 15 \text{ or } x = 15 (\text{as } AP < AB)$$

$$\text{Or } PB = AB - AP = 20 - 15 = 5 \text{ km}$$

70. A

Sol:



$$AD = AB \cos \theta = 2R \cos \theta$$

$$AE = AD \cos \theta = 2R \cos^2 \theta$$

$$\text{Or } EF = AB - 2AE = 2R - 4R \cos^2 \theta$$

$$DE = AD \sin \theta = 2R \sin \theta \cos \theta$$

Thus, area of trapezium,

$$S = \frac{1}{2}(AB + CD) \times DE$$

$$= \frac{1}{2}(2R + 2R - 4R \cos^2 \theta) \times 2R \sin \theta \cos \theta$$

$$= 4R^2 \sin^3 \cos \theta$$

$$\frac{dS}{d\theta} = 12R^2 \sin^2 \theta \cos^2 \theta - 4R^2 \sin^4 \theta$$

$$= 4R^2 \sin^2 \theta (3 \cos^2 \theta - \sin^2 \theta)$$

$$\text{For maximum area, } \frac{dS}{d\theta} = 0 \text{ or } \tan^2 \theta = 3$$

$$\text{Or } \tan \theta = \sqrt{3} (\theta \text{ is acute}) \text{ or } S_{\max} = \frac{3\sqrt{3}}{4} R^2$$

71. A

$$\text{Sol: } u = \sqrt{c+1} - \sqrt{c}$$

$$u = \frac{1}{\sqrt{c+1} + \sqrt{c}} \text{ and } v = \frac{1}{\sqrt{c-1} + \sqrt{c}}$$

Clearly, $u < v$.

Also, f is increasing whereas g is decreasing. Thus,

$$u < v$$

$$\text{Or } f(u) < f(v)$$

$$\text{Or } \text{gof}(u) > \text{gof}(v)$$

72. D

73. B

Sol: Let the polynomial be $P(x) = ax^2 + bx + c$

$$\text{Given } P(0) = 0 \text{ and } P(1) = 1$$

$$\Rightarrow c = 0 \text{ and } a + b = 1 \Rightarrow a = 1 - b$$

$$\therefore P(x) = (1 - b)x^2 + bx, \Rightarrow P'(x) = 2(1 - b)x + b$$

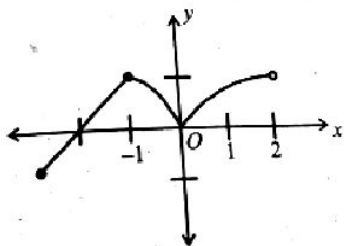
$$\text{Given } P'(x) > 0, \forall x \in [0, 1] \Rightarrow 2(1 - b)x + b > 0$$

$$\Rightarrow \text{When } x = 0, b > 0 \text{ and when } x = 1, b < 2 \Rightarrow 0 < b < 2$$

$$\text{Hence, } S = \{(1 - a)x^2 + ax, a \in (0, 2)\}$$

74. C

Sol: Given function: $f(x) = \begin{cases} (2+x)^3, & -3 < x \leq -1 \\ x^{2/3}, & -1 < x < 2 \end{cases}$



The graph of $y = f(x)$ is as shown in the figure. From graph, clearly there is one local maximum at $x = -1$ and one local minima at $x = 0$

$\therefore$ Total number of local maxima or minima = 2.

75. A

Sol: Given: $f(x) = x^2$ and $g(x) = \sin x, \forall x \in \mathbb{R}$

$$\therefore (\text{gof})(x) = \sin x^2$$

$$\Rightarrow (\text{gogof})(x) = \sin(\sin x^2)$$

$$\Rightarrow (\text{fogogof})(x) = \sin^2(\sin x^2)$$

Since given that $(\text{fogogof})(x) = (\text{gogof})(x)$

$$\therefore \sin^2(\sin x^2) = \sin(\sin x^2)$$

$$\Rightarrow \sin(\sin x^2) = 0, 1$$

$$\Rightarrow \sin x^2 = n\pi \text{ or } ((4n+1)\frac{\pi}{2}), \text{ where } n \in \mathbb{Z}$$

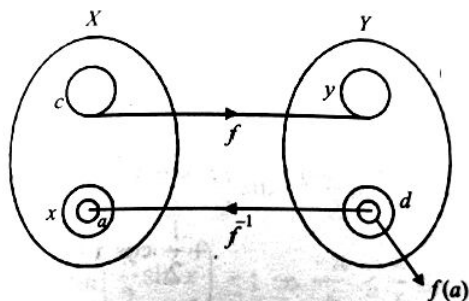
$$\Rightarrow \sin x^2 = 0 \quad (\because \sin x^2 \in [-1, 1]) \Rightarrow x^2 = n\pi$$

$$\therefore x = \pm\sqrt{n\pi}, \text{ where } n \in \mathbb{W}$$

76. D

Sol: Given that X and Y are two sets and $f: X \rightarrow Y$.

$\{f(c) = y; c \in X, y \in Y\}$ and $\{f^{-1}(d) = x; d \in Y, x \in X\}$ The pictorial representation of given information is as shown.



Since $f^{-1}(D) = x \Rightarrow f(x) = d$

Now if $a \in x \Rightarrow f(a) \in f(x) = d \Rightarrow f^{-1}(f(a)) = a$

Hence, $f^{-1}(f(a)) = a, a \in x$ is the correct option.

77. C

Sol: Given that $f: A \rightarrow B$ and $g: B \rightarrow C$

$\therefore f^{-1}: B \rightarrow A$ and $g^{-1}: C \rightarrow B$

We have $(g \circ f)^{-1} = f^{-1} \circ g^{-1}: C \rightarrow A$

$\therefore f$ must be one - one and g will be onto function

PASSAGE-1:

78. 3

79. 4

80. 3

Sol. 78, 79, 80

We have $A_i = \frac{x - a_i}{|x - a_i|}, i = 1, 2, \dots, n,$

And $a_1 < a_2 < \dots < a_{n-1} < a_n$.

Let x be in the left neighbourhood of a_m .

Then $x - a_i < 0$ for $i = m, m+1, \dots, n$

And $x - a_i > 0$ for $i = 1, 2, \dots, m-1$

And $A_i = \frac{x - a_i}{-(x - a_i)} = -1$ for $i = m, m+1, \dots, n$

$A_i = \frac{x - a_i}{x - a_i} = 1$ for $i = 1, 2, \dots, m-1$

Similarly, if x is in the right neighbourhood of a_m , then $x - a_i < 0$ for $i = m+1, \dots, n$, and $x - a_i > 0$ for $i = 1, 2, \dots, m$.

Therefore,

$A_i = \frac{x - a_i}{-(x - a_i)} = -1$ for $i = m+1, \dots, n$

$$\lim_{x \rightarrow a_m} (A_1 A_2 \dots A_n) = (-1)^{n-m+1}$$

$$\text{And } \lim_{x \rightarrow a_m^+} (A_1 A_2 \dots A_n) = (-1)^{n-m}$$

Hence, $\lim_{x \rightarrow a_m} (A_1 A_2 \dots A_n)$ does not exist.

PASSAGE-2:

81. 2

82. 4

83. 3

Sol: 81,82,83

$$\begin{aligned} L &= \lim_{x \rightarrow 0} \frac{\sin x + ae^x + be^{-x} + c \log_e(1+x)}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{\left(x - \frac{x^3}{3!}\right) + a\left(1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!}\right) + b\left(1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!}\right) + c\left(x - \frac{x^2}{2} + \frac{x^3}{3}\right)}{x^3} \\ &= \lim_{x \rightarrow 0} \frac{(a+b) + (1+a-b+c)x + \left(\frac{a}{2} + \frac{b}{2} - \frac{c}{2}\right)x^2 + \left(-\frac{1}{3!} + \frac{a}{3!} - \frac{b}{3!} + \frac{c}{3}\right)x^3}{x^3} \\ &\Rightarrow a+b=0, 1+a-b+c=0, \frac{a}{2} + \frac{b}{2} - \frac{c}{2} = 0 \end{aligned}$$

$$\text{And } L = -\frac{1}{3!} + \frac{a}{3!} - \frac{b}{3!} + \frac{c}{3!}$$

Solving first three equations we get $c=0, a=-1/2, b=1/2$.

$$\therefore L = -1/3$$

Equation $ax^2 + bx + c = 0$ reduces to $x^2 - x = 0 \Rightarrow x = 0, 1$

$\|x+c|-2a| < 4b$ reduces to $\|x|+1| < 2$

$$\Rightarrow -2 < |x|+1 < 2$$

$$\Rightarrow 0 \leq |x| < 1$$

$$\Rightarrow x \in [-1, 1]$$

PASSAGE-3:

84. 1

85. 3

86. 4

Sol: 84, 85, 86

$$\begin{aligned} F(x) &= \lim_{n \rightarrow \infty} \frac{f(x) + x^{2n}g(x)}{1 + x^{2n}} \\ &= \begin{cases} f(x), & 0 \leq x^2 < 1 \\ \frac{f(x)+g(x)}{2}, & x^2 = 1 \\ g(x), & x^2 > 1 \end{cases} \end{aligned}$$

$$= \begin{cases} g(x), & x < -1 \\ \frac{f(-1)+g(-1)}{2}, & x = -1 \\ f(x), & -1 < x < 1 \\ \frac{f(1)+g(1)}{2}, & x = 1 \\ g(x), & x > 1 \end{cases}$$

For continuity at $x = -1$

$$f(-1) = g(-1) \text{ or } 1 - a + 3 = b - 1 \text{ or } a + b = 5$$

For continuity at $x = 1$,

$$f(1) = g(1) \text{ or } 1 + a + 3 = 1 + b \text{ or } a - b = -3$$

Solving equations (1) and (2), we get $a = 1$ and $b = 4$.

$$f(x) = 2g(x) \Rightarrow x^2 + x + 3 = 2x + 8 \text{ or } x^2 - x - 5 = 0$$

PASSAGE-4:

87. 2

88. 3

89. 1

Sol: 87, 88, 89

$$\text{Given } \lim_{x \rightarrow \pm\infty} f(x) = \infty$$

$\therefore$ degree of $g(x)$ must be less than 4.

$$\text{Also } \lim_{x \rightarrow -1} f(x) = 6$$

$$\therefore \lim_{x \rightarrow -1} \frac{(x^2 - 1)(x^2 - 4)}{g(x)} = 6$$

$\therefore g(x)$ must have $(x + 1)$ as a factor.

$$\text{Also } \lim_{x \rightarrow -2} f(x) \neq 0 \text{ but finite}$$

$\Rightarrow g(x)$ must have $(x + 2)$ another factor.

$$\Rightarrow g(x) = k(x + 1)(x + 2)$$

$$\Rightarrow \lim_{x \rightarrow -1} \frac{(x^2 - 1)(x^2 - 4)}{k(x + 1)(x + 2)} = 6$$

$$\Rightarrow \lim_{x \rightarrow -1} \frac{(x - 1)(x - 2)}{k} = 6$$

$$\text{Thus } f(x) = \frac{(x^2 - 1)(x^2 - 4)}{g(x)}$$

$$= \frac{(x^2 - 1)(x^2 - 4)}{(x + 1)(x + 2)}, x \neq -1, -2$$

$$= (x - 1)(x - 2), x \neq -1, -2$$

$|f(x)| = |(x - 1)(x - 2)|$ is non-differentiable at $x = 1$ and $x = 2$

$f(|x|) = (|x| - 1)(|x| - 2)$ is non-differentiable at $|x| = 1, 2$ and $x = 0$

Or $x = \pm 1, \pm 2, 0$

$f(-|x|) = (-|x| - 1)(-|x| - 2) = (|x| + 1)(|x| + 2)$ is non-differentiable at $x = 0$ only.

90. 4

91. 4

92. 2

Sol: 90, 91, 92

$$x = f(t) = a^{\ln(b^t)} = a^{t \ln b} \quad \dots(1)$$

$$y = g(t) = b^{-\ln(a^t)} = (b^{\ln a})^{-t} = (a^{\ln b})^{-t} = a^{-t \ln b}$$

$$\therefore y = g(t) = a^{\ln(b^{-t})} = f(-t) \quad \dots(2)$$

From equations (1) and (2),

$$xy = 1$$

90. $y = \frac{1}{x}$

$$\therefore \frac{dy}{dx} = -\frac{1}{x^2} = -\frac{1}{f^2(t)}$$

$$\text{Also, } xy = 1 \Rightarrow -\frac{1}{f^2(t)} = g^2(t)$$

$$\therefore \frac{dy}{dx} = -\frac{1}{x^2} = -\frac{y^2}{1}$$

$$\text{Also, } xy = 1 \Rightarrow \frac{dy}{dx} = -\frac{y}{x} = -\frac{g(t)}{f(t)}$$

91. $f(t) = g(t) \Rightarrow f(t) = f(-t) \rightarrow t = 0$

$[\because f(t) \text{ is one - one function}]$

At $t = 0, x = y = 1$

$$\therefore xy = 1, \frac{dy}{dx} = -\frac{1}{x^2} \text{ and } \frac{d^2y}{dx^2} = \frac{2}{x^3} \text{ at } x = 1, \frac{d^2y}{dx^2} = 2$$

92. $\therefore xy = 1$

$$fg = 1$$

$$\therefore fg' + gf' = 0$$

$$\therefore fg'' + g'f' + g'f' + gf'' = 0$$

$$\text{Or } fg'' + gf'' + 2g'f' = 0$$

$$\text{Or } \frac{f}{f'} \frac{g''}{g'} + \frac{gf''}{g'f'} = -2$$

From equation (2), $g(t) = f(-t)$

$$\therefore g'(t) = -f'(-t)$$

$$\text{And } g''(t) = f''(-t)$$

Substituting in equation (3), we get

$$\frac{f(t)}{f'(t)} \cdot \frac{f''(-t)}{-f'(-t)} + \frac{f(-t)}{-f'(-t)} \cdot \frac{f''(t)}{f'(t)} = -2$$

$$\frac{f(t)}{f'(t)} \cdot \frac{f''(-t)}{f'(-t)} + \frac{f(-t)}{f'(-t)} \cdot \frac{f''(t)}{f'(t)} = 2$$

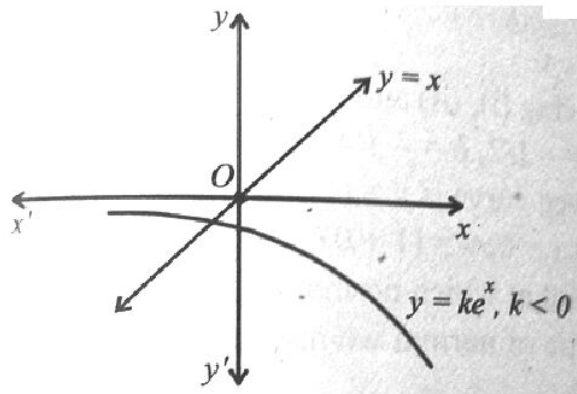
93.

93. C

94. A

95. A

Sol: 93, 94, 95

93 For $k = 0$, line $y = x$ meets $y = 0$, i.e., x -axis only at one point.For $k < 0$, $y = ke^x$ meets $y = x$ only once as shown in the graph.94. Let $f(x) = ke^x - x$ Now for $f(x) = 0$ to have only one root means the line $y = x$ must be tangential to the curve $y = ke^x$ Let it be so at (x_1, y_1) , then

$$\left(\frac{dy}{dx}\right)_{\text{curve 1}} = \left(\frac{dy}{dx}\right)_{\text{curve 2}} \Rightarrow 1 = ke^{x_1} \Rightarrow e^{x_1} = \frac{1}{k}$$

Also

$$y_1 = ke^{x_1} \text{ and } y_1 = x_1$$

$$\Rightarrow x_1 = 1 \Rightarrow 1 = ke \Rightarrow k = 1/e$$

95. For $y = x$ to be tangent to the curve

$$y = ke^x, k = 1/e$$

 $\therefore$ for $y = ke^x$ to meet $y = x$ at two points we should have

$$k < \frac{1}{e} \Rightarrow k \in \left(0, \frac{1}{e}\right) \text{ as } k > 0.$$

96.

96. 4

97. 4

98. 2

Sol: $f(x) = x + \cos x - a$ or $f'(x) = 1 - \sin x \geq 0 \forall x \in \mathbb{R}$.Thus, $f(x)$ is increasing in $(-\infty, \infty)$ as for $f'(x) = 0$, x is not forming an interval. Also,

$$f''(x) = -\cos x = 0$$

Or $x = (2n+1)\frac{\pi}{2}, n \in \mathbb{Z}$

Hence, there are infinite points of inflection.

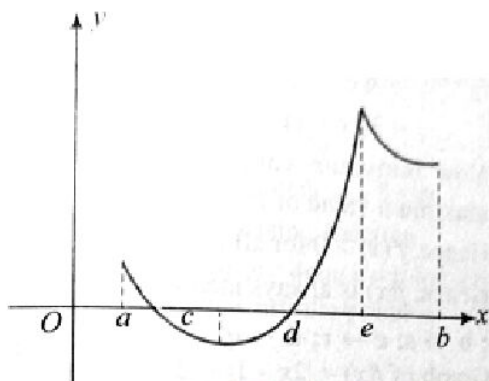
Now, $f(0) = 1 - a$

For positive root, $1 - a < 0$ or $a > 1$. For negative root,

$1 - a > 0$ or $a < 1$.

99.

99.sol.



$f'(x)$ changes sign from positive to negative at $x = c$.

So $x = c$ is point maxima.

$f'(x)$ changes sign from negative to positive at $x = d$.

So $x = d$ is point minima

100. Clearly one point of inflection lies in the interval (a, d) where derivative of $f'(x)$ is zero.

Also $f''(x)$ does not exist at $x = e$, but $f''(x)$ changes sign in the neighbourhood of e , so, $x = e$ is another point of inflection.

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